Multimodal Fusion Transformer for Remote Sensing Image Classification

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***Abstract*— Vision transformers (ViTs) have been trending in image classification tasks due to their promising performance when compared with convolutional neural networks (CNNs). As a result, many researchers have tried to incorporate ViTs in hyperspectral image (HSI) classification tasks. To achieve satisfactory performance, close to that of CNNs, transformers need fewer parameters. ViTs and other similar transformers use an external classification (`CLS`) token, which is randomly initialized and often fails to generalize well, whereas other sources of multimodal datasets, such as light detection and ranging (LiDAR), offer the potential to improve these models by means of a `CLS`. In this article, we introduce a new multimodal fusion transformer (MFT) network, which comprises a multihead cross-patch attention (`mCrossPA`) for HSI land-cover classification. Our `mCrossPA` utilizes other sources of complementary information in addition to the HSI in the transformer encoder to achieve better generalization. The concept of tokenization is used to generate `CLS` and HSI patch tokens, helping to learn a distinctive representation in a reduced and hierarchical feature space. Extensive experiments are carried out on widely used benchmark datasets, i.e., the University of Houston (UH), Trento, University of Southern Mississippi Gulfpark (MUUFL), and Augsburg. We compare the results of the proposed MFT model with other state-of-the-art transformers, classical CNNs, and conventional classifiers models. The superior performance achieved by the proposed model is due to the use of `mCrossPA`. The source code will be made available publicly at https://github.com/AnkurDeria/MFT.**

***Index Terms*— Convolutional neural networks (CNNs), multihead cross-patch attention (`mCrossPA`), remote sensing (RS), vision transformer (ViT).**

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I. Introduction and Contributions

PHENOMENA, such as climate change or desertification, have led to a drastic growth in the popularity of Earth observation (EO) via remote sensing (RS). These tasks include (but are not limited to) land-cover classification [1], [2], [3], forestry [4], mineral exploration and mapping, object/target detection [5], [6], environmental monitoring [7], urban planning [8], biodiversity conservation, and disaster response and management. All of these tasks have been explored in the past few decades using data coming from single EO sensors, i.e., hyperspectral imaging (HSI) instruments, which can simultaneously provide rich spectral and spatial information [9]. However, such single-sensor data tend not sufficient to identify and recognize objects of interest.

Recent advances in RS technology have increased the availability of multisensor data, allowing for multiple representations of the same geographical region. Depending on the sensors' characteristics, the captured data can provide information with different characteristics for the same observed land-cover region. For example, synthetic-aperture radar (SAR) data provide the amplitude and phase geometrical information, while light detection and ranging (LiDAR) collects depth and intensity information, measuring the elevation of an object on the Earth's surface. Multispectral (MS) sensors measure the reflected light within specific wavelength ranges across the electromagnetic spectrum. Integrating these data with different modalities provides unique complementary information, allowing us to further obtain complete feature representations [10].

However, from the perspective of an RS imaging system, the compatibility of spatial and spectral resolution is often questionable [11]. The very high spectral resolution of HSI data often limits spatial resolution. As a consequence, spectral aliasing becomes a phenomenon that adversely affects land-cover classification in complex scenes [12]. In contrast, LiDAR provides elevation information that allows distinguishing land-cover objects with identical spectral signatures but different elevations, such as roads and roofs with built-in cement. For instance, HSI data can accurately differentiate between water and grass. But, for instance, roads and roofs built using the same materials cannot be differentiated. Therefore, the complementary elevation information obtained from LiDAR can be greatly beneficial to classification purposes [13]. Khodadadzadeh et al. [14] extracted geometric representations from LiDAR data for land-cover classification.

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Many works have explored to fully exploit the complementary information between HSI and other sources of multimodal data, such as LiDAR, SAR, MS, or digital surface models (DSMs) [10], [15]. In this context, researchers used extended morphological profiles (EPs) and attribute profiles (APs) for joint feature extraction and classification of HSI and LiDAR data [12], [16], [17]. Rasti et al. [18] improved the joint extraction of EPs by applying total variation component analysis for feature fusion. Merentitis et al. [19] introduced an automatic fusion technique using the random forest (RF) classifier for joint HSI and LiDAR data classification.

To adequately exploit the information in both HSI and other sources of multimodal data, such as SAR, DSM, and LiDAR, we propose a new yet efficient multimodal fusion transformer (MFT) network, which uses multihead cross-patch attention (`mCrossPA`) mechanism to fuse other sources of complementary information and HSI patch tokens for land-cover classification. Our newly developed MFT incorporates the best properties of transformers encoder and `mCrossPA` for the fusion of RS classification tasks without introducing much computational overhead. The major contributions of this article can be summarized as follows.

1) We propose a new MFT network to improve RS data fusion for land-cover classification, where HSIs are used along with other multimodal data sources (e.g., LiDAR, MS image (MSI), SAR, and DSM) in a transformer network to enhance the classification performance.
2) We introduce a novel `mCrossPA` mechanism for RS data fusion. The class token, which also contains supplementary information, is derived from multimodal data (e.g., LiDAR, MSI, SAR, and DSM), which are fed to the transformer network along with HSI patch tokens.
3) The newly developed `mCrossPA` utilizes the widely used attention mechanism that can efficiently fuse the information from HSI patch tokens and existing external classification (`CLS`) tokens into a new token that integrates multimodal features.
4) We conduct extensive experiments on four public hyperspectral datasets, i.e., Houston, Trento, University of Southern Mississippi Gulfpark (MUUFL), and Augsburg. These experiments reveal the effectiveness of the proposed method. To illustrate the advantages of our approach, we use HSI data alone and also combine it with other sources of multimodal data.

The remaining of this article is organized as follows. Section II discusses the traditional methods, classic deep learning (DL)-based methods, and transformer-based methods used for HSI classification. The preprocessing mechanisms for HSI and other sources of multimodal data and the components of the proposed MFT model, including a novel LiDAR and HSI CrossPA mechanism for transformer-based deep feature fusion and image classification, are described in Section III. Extensive experiments are then conducted, with an analysis of hyperparameter sensitivity and a discussion of the obtained results in Section IV. This article concludes with some remarks and hints at plausible future research lines in Section V.

II. Related Works

A. Traditional Methods

Conventional methods have been widely used for HSI classification, even with limited training samples [20], [21], [22]. Generally speaking, these methods incorporate two steps. First, they represent the HSI data in feature space to reduce the dimensionality and extract a few highly informative features. Then, the extracted features are sent to a spectral classifier [1], [23], [24], [25], [26]. Among conventional methods, support vector machines (SVMs) with nonlinear kernels are very popular, especially when the training data are limited [27]. The extreme learning machine (ELM) has also been widely used for unbalanced feature classification, and it was used in HSI classification. It has been shown that ELMs were able to achieve better performance when compared with SVMs. The RF has been used due to its discriminative power when classifying unlabeled HSIs [28]. However, conventional methods encounter a performance bottleneck when the training data becomes complex, due to their limitations in terms of data fitting and representation ability. The aforementioned spectral classifiers consider HSIs as a collection of spectral measurements without considering their spatial arrangement. Spatial–spectral classifiers considerably boost the performance of spectral classifiers by incorporating spatial information, such as the shape and size of different objects and textures. It is also worth mentioning that the spectral classifiers are less robust to noise compared with the spatial–spectral classifiers [1], [29].

B. Conventional DL Methods

In recent years, multimodal data integration for RS data classification using DL methods has attracted significant attention [30]. DL methods can efficiently learn spectral and spatial features from vast amounts of fragmented data. In the case of HSI-LiDAR data integration, both supervised and unsupervised DL methods have shown efficient performance in deep feature representation. Hong et al. [31] developed a new technique for classifying multisource RS data (HSI and LiDAR), which can effectively extract compact feature representations of multimodal RS data. Very recently, Hong et al. [32] first proposed a general and unified multimodal DL framework for RS image classification, which provides an effective solution with respect to multisource and multimodal RS products.

Unlike unsupervised methods, which learn the feature representation using observed data, supervised methods rely on ground reference training samples. Therefore, DL-based supervised techniques often demonstrate superiority in RS image classification tasks. One-dimensional convolution (CNN1D) [33], two-dimensional convolution (CNN2D) [34], and three-dimensional convolution (CNN3D) [35] have shown success in HSI and LiDAR data classification. Recent studies show that DL methods can balance algorithm accuracy and robustness. On the other hand, shallow learning methods are dependent on the prior information obtained from training or observed data. He et al. [36] proposed a residual network (ResNet) with minimum information loss after

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each convolutional operation to redeem the vanishing gradient problem. Zhong et al. [37] designed a spectral–spatial residual network (SSRN) that can better utilize both spectral and spatial information for enhanced classification. Roy et al. [38], [39] explored the light-weighted paradigm by modeling spectral and spatial features extracted through the squeeze-and-excitation ResNet, which can be combined with bag-of-features learning for accurate land-use and land-cover classification. To boost the performance of SSRNs, Zhu et al. [40] included another spatial and channel attention layer in the SSRN architecture to extract discriminative representation. To take advantage of the residual behavior of the network, one can extend it to form an even more complex model, i.e., the lightweight spectral–spatial squeeze-and-excitation attention [38] with adaptive kernels [20] and pyramidal ResNets [41]. Lightweight heterogeneous kernel convolution [42], rotation equivariant convolutional neural network (CNN) [43], and gradient centralized convolution [3], [44] enable efficient feature extraction and classification. On the other hand, the generative adversarial network also provides a solution for the classification of HSIs containing a classwise imbalanced number of samples [45], [46].

The networks mentioned above (and their variants) are generally insufficient for detecting subtle discrepancies among spectral dimensions. CNNs, which have shown their capacity in the task of extracting spatial–contextual information from HSIs, barely capture the sequence attributes, especially the middle- and long-term dependencies. That leads to a decrease in performance, especially when the data comprise many classes with similar spectral signatures, creating problems in extracting diagnostic spectral attributes. On the other hand, recursive neural networks (RNNs) can accurately model spectral signatures from HSIs by accumulating them band by band in an orderly manner. As such, the order of spectral bands is essential in learning long-term dependencies and preventing gradient vanishing problems [47], which might further complicate the interpretation of spectrally salient changes. This is because HSIs contain limited samples, and RNNs cannot train the model concurrently, which limits the classification performance. In this work, we rethink HSI data classification using transformers to address the aforementioned limitations.

C. Transformer-Based Methods

Unlike CNNs and RNNs, transformers are one of the most cutting-edge backbone networks due to their adoption of self-attention techniques, which are efficient for processing and analyzing sequential (or time series) data [48]. Inspired by the success of transformers in computer vision [49], several new transformer models have been developed in the last few years [50]. Although the transformer excels at capturing the information containing in spectral signatures, it cannot equally characterize local semantic elements and fails to make adequate use of spatial information. The original transformer [51] is a model based on the self-attention mechanism that is mainly used in natural language processing (NLP). A series of tokens are used as model's input, and multihead attention is employed to draw global correlations in the input character sequence. Hong et al. [22] recently introduced SpectralFormer, which can learn spectral representation information using neighboring bands and can construct a cross-layer transformer encoder module. The spectral–spatial feature tokenization (SSFT) transformer model [52] was suggested to make use of the transformer's capacity to extract local spatial semantic information and represent the link between neighboring sequences. The major drawback of this model is the number of parameters that are used. SSFT performs well with HSI data; however, it cannot include other multimodal data sources during the classification task. Ding et al. [53] introduced a global–local transformer network (GLT-Net) to fully exploit the potential of transformer architectures for modeling the long-range dependencies as well as the convolution operator, which characterizes locally correlated features in joint classification of HSI and LiDAR data. To fuse multisource heterogeneous information and enhance the joint classification performance, a novel dual-branch network was presented by Zhao et al. [54] that consists of a transformer network and a hierarchical CNN. Yu et al. [55] created a cross-context capsule vision transformer (CapViT) to take advantage of long-range global feature interactions at various context scales for land-cover classification with MS-LiDAR data. In order to capture the relationship between HSI and its complementary information, such as LiDAR, as well as to fuse the multisource features and prevent the generation of redundant information, the local information interaction transformer (LIIT) model was introduced for land-cover classification using a combination of HSI and LiDAR data [56]. A multiattentive hierarchical fusion network (MAHiDFNet) was also developed in [30] to implement feature-level fusion and classification of HSIs using LiDAR data. The connection between the structure and shape information of the HSI token and the CLS token was enhanced via spectral and spatial morphological convolution procedures in conjunction with a self-attention mechanism (morphFormer) for land-cover classification in [57].

III. Methodology

A. Preprocessing of HSI and LiDAR Data

Let us denote an HSI as $\mathbf{X_H} \in \mathcal{R}^{M\times N\times B}$ and a corresponding LiDAR image as $\mathbf{X_L} \in \mathcal{R}^{M\times N}$, where M and N, respectively, represent two spatial dimensions and B refers to the number of spectral bands of the HSI. All the pixels are classified into C land-cover classes denoted by $C = (y_1, y_2, \ldots y_c)$. A single pixel can be defined as $\mathbf{x}_{i,j} \in \mathbf{X_H}$, where $i = 1, \ldots M$ and $j = 1, \ldots N$ with $\mathbf{x}_{i,j} = [x_{i,j,1}, \ldots x_{i,j,B}] \in \mathcal{R}^B$ containing B spectral bands. Patch extraction is performed in a preprocessing step, where a spectral–spatial cube $\mathbf{x}_h^{i,j} \in \mathcal{R}^{k\times k\times B}$ is obtained from the normalized HSI data $\mathbf{X_H}$ and a spatial image $\mathbf{x}_l^{i,j} \in \mathcal{R}^{k\times k}$ is obtained from the LiDAR data $\mathbf{X_L}$, with neighboring regions of size $(k \times k)$ and centered at pixel (i, j).

By jointly exploiting the spectral–spatial information, we can increase the discriminative power of the feature learning network. For this reason, spectral–spatial cubes $\mathbf{x}_h^{(i,j)}$ are extracted from the original HSIs and stacked into X_H prior to the feature extraction process. Similarly, for the LiDAR data,

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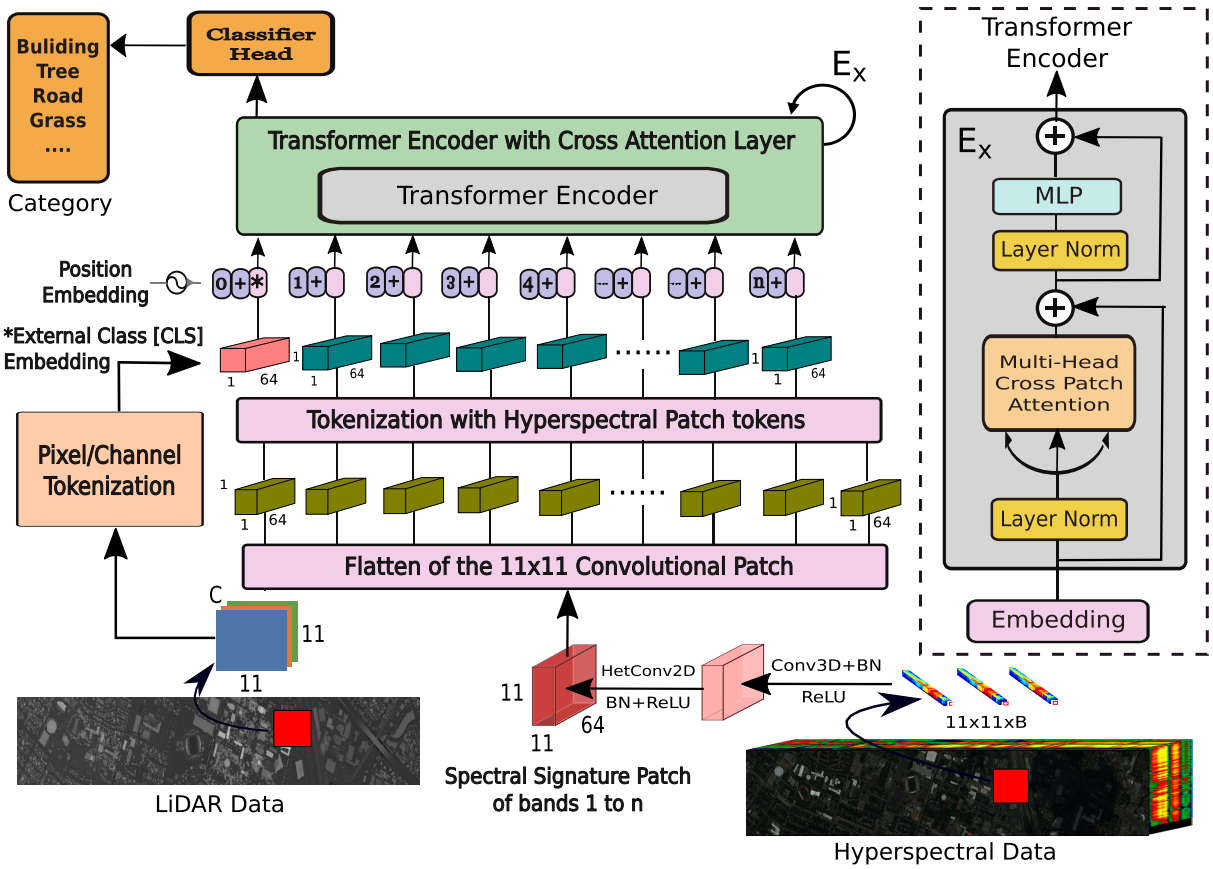


Fig. 1. Graphical representation of the proposed MFT network for HSI and LiDAR data fusion.

spatial patches of the same size $\mathbf{x}_l^{(i,j)}$ are extracted and stacked into X_L. Finally, the training and test samples in each class for X_H and X_L can be represented by

$$D^{\text{train}} = \{(x_h, x_l), y^{(i)} | i = 1, \ldots P\}$$
$$D^{\text{test}} = \{(x_h, x_l), y^{(i)} | i = 1, \ldots Q\} \quad (1)$$

where $x_h \in X_H$ and $x_l \in X_L$ are randomly chosen from the HSI and LiDAR data, P and Q represent the number of training and test samples, and $y^{(i)}$ is the actual class for the ith land-cover pixel.

B. Proposed MFT

Fusion is crucial for effectively learning multimodal feature representations of RS data, since complementary information of a model can enhance the learning of discriminative features. If we incorporate this technique into conventional transformer models, we may exponentially increase the number of parameters. For instance, in conventional ViT models, if we use the HSI data as input, we increase the linear projection's complexity due to the fact that the input has numerous spectral bands, which may lead to chances of overfitting. If we assume that other multimodal data (e.g., LiDAR, SAR, or DSM) are concatenated to the HSIs to learn complementary information, the aforementioned problem is aggravated due to the increase in the number of bands. Hence, a transformer model can effectively learn multimodal information without increasing computational overhead of linear projection.

To address the aforementioned challenges, we introduce an MFT where the input patches are taken from the HSIs. In addition, the `CLS` token is generated from a patch that describes the same spatial region as the HSI patch, taken from the corresponding LiDAR image (or other modalities, such as SAR, MSI, DSM, and so on). In addition, we propose a simple yet effective `mCrossPA` module to fuse the LiDAR (as `CLS`) token and the HSI patch tokens. Fig. 1 depicts the proposed MFT network architecture for multimodal RS image fusion to improve the land-cover classification performance. For the sake of simplicity, only LiDAR data from the University of Houston (UH) dataset are considered, but the concept can be extended to other multimodal data as well. The results for different kinds of multimodal data (i.e., MS, SAR, and DSM) are reported in the results section. The objective of the proposed MFT model is to learn the spectral–spatial patch embeddings instead of the bandwise embeddings of the input HSIs, in addition to enriching the abstract description of the `CLS` token by considering LiDAR as an external class embedding without introducing any computational overhead. The components of the proposed MFT are discussed step by step as follows.

1) HSI and LiDAR Feature Learning via CNNs: The ability of automatic contextual modeling among features helps CNNs to make strong inferences and exhibit promising performance

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in HSI classification tasks. The availability of a large number of spectral bands in HSIs enables us to utilize the benefits of CNNs and control the depth of the output feature maps. It has already been shown that CNNs can extract high-level abstract features, which are also invariant to the source of data modality. Hence, the CNNs adopted in our proposed model extract high-level abstract features that are used as input in the transformer and reduce the HSI bands down to a suitable number.

In order to extract robust and discriminative features from raw HSIs, sequential layers of Conv3D and `HetConv2D` are used. Fig. 1 shows a graphical representation of the `Conv3D` and `HetConv2D` network for HSIs. The HSI cube $\mathbf{X}_H$ of size $(11 \times 11 \times B)$ is first unsqueezed into shape $(1 \times 11 \times 11 \times B)$, passed through the first `Conv3D` layer to produce $X_{in} = \text{Conv3D}(X_H)$, and the size of the considered kernels of that layer is chosen to be $(3 \times 3 \times 9)$, while the padding is $(1 \times 1 \times 0)$ to keep the spatial height and width of the output image equal to those of the input. On the other hand, the `HetConv2D` [58] utilizes two `Conv2D` layers in parallel, one of which performs a groupwise convolution (with kernel size $= 3$, groups $= 4$, and padding $= 1$) and the other one performs a pointwise convolution (with kernel size $= 1$, groups $= 1$, and padding $= 0$). The outputs from both these convolutions are added in elementwise ($\oplus$) fashion and returned from the `HetConv2D` layer. These can be defined as follows:

$$X_{\text{out}} = \begin{cases} \text{Conv2D}(X_{\text{in}}, k = (3,3), s = 1, g = 4, p = 1) \\ \oplus \\ \text{Conv2D}(X_{\text{in}}, k = (1,1), s = 1, g = 1, p = 1). \end{cases} \tag{2}$$

The shapes of the output feature maps after the `Conv3D` layer and the `HetConv2D` layer are $[8 \times 11 \times 11 \times (B-8)]$ and $(11 \times 11 \times 64)$, respectively. Both of these layers are followed by their respective batch normalization (BN) [59] and rectified linear unit (ReLU) activation layers. The BN layer is used to address the issue of overfitting (due to the small number of training samples) and also to accelerate the training performance. ReLU plays an important role by introducing nonlinearities into the output features maps in order to achieve efficient and smooth propagation of the loss gradient.

2) HSI and LiDAR Tokenization: Spectral and spatial features contain highly discriminative information, leading to higher accuracy in HSI classification tasks. HSI cubes of size $\{(11 \times 11) \times 64\}$ are flattened into feature patch tokens with shape (1×64) each. n such patches are selected from 121 patches using tokenization [52]. The tokenization operation uses two learnable weights, i.e., W_a and W_b, which are multiplied by the input to extract the key features. The following equation summarizes the process:

$$\begin{aligned} X^H_{\text{flat}} &= T(\text{Flatten}(X_{\text{out}})) \\ X^H_{W_a} &= \text{softmax}\left(T\left(X^H_{flat}.W_{aH}\right)\right) \\ X^H_{W_b} &= X^H_{flat}.W_{bH} \\ X^H_{\text{patch}} &= X^H_{W_a}.X^H_{W_b} \end{aligned} \tag{3}$$

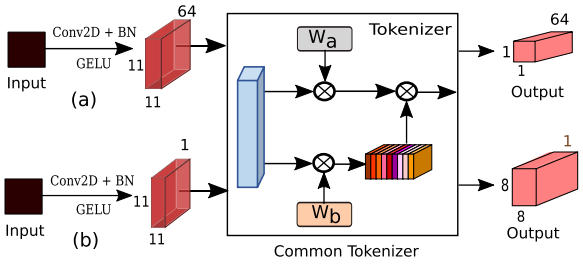


Fig. 2. Graphical representation of (a) HT and (b) PT.

where $T(.)$ is a transpose function, $X^H_{\text{flat}} \in \mathcal{R}^{121\times 64}$, $W_a \in \mathcal{R}^{64\times n}$ and $W_b \in \mathcal{R}^{64\times 64}$, $X^H_{W_a} \in \mathcal{R}^{n\times 121}$ and $X^H_{W_b} \in \mathcal{R}^{121\times 64}$, and $X^H_{\text{patch}} \in \mathcal{R}^{n\times 64}$.

Focusing on the LiDAR data, a patch of size $(11 \times 11 \times C)$ extracted from a LiDAR image is first passed through either a pixel tokenization or a channel tokenization operation shown in Fig. 1. Both operations output a `CLS` token of shape (1×64) generated from the input LiDAR patch by passing it through a `Conv2D` layer, followed by a BN layer, a `GELU` activation function, and a tokenizer, respectively. Fig. 2 illustrates the difference between the pixel and the channel tokenization operations. The `Conv2D` layer may reduce the number of channels to 1 or increase it to 64 depending on the variation used. The convolutional layer extracts different spatial elevation information of height and shape, such as roads and roof built-in cement having different heights, which facilitates the exchange of complementary information among the HSI patch tokens (X^H_{patch}). The LiDAR data may have only one channel to begin with, but other multimodal types of data, such as SAR, DSM, MSI, and so on, may have several channels, which need to be reduced to one, since the aim of the network is to use the LiDAR data as the `CLS` token, which should have only one channel. The input and output spatial heights and widths are kept the same through the convolutional layer using a kernel of size 3 and padding of size 1. The LiDAR feature embeddings are now suitable to be used as the `CLS` token. The whole step can be summarized as follows:

$$\begin{aligned} X^L_{\text{conv}} &= T\left(\text{Flatten}\left(\text{Conv2D}\left(X^L_{\text{patch}}\right)\right)\right) \\ X^L_{W_a} &= \text{softmax}\left(T\left(X^L_{\text{conv}}.W_{aL}\right)\right) \\ X^L_{W_b} &= X^L_{\text{conv}}.W_{bL} \\ X^L_{\text{cls}} &= X^L_{W_a}.X^L_{W_b} \end{aligned} \tag{4}$$

where $X^L_{\text{patch}} \in \mathcal{R}^{11\times 11\times C}$, $T(.)$ is a transpose function, and, in the case of the pixel tokenizer (PT), $X^L_{\text{conv}} \in \mathcal{R}^{121\times 1}$, $W_{aL} \in \mathcal{R}^{1\times 1}$, and $W_{bL} \in \mathcal{R}^{1\times 64}$. In the case of the channel tokenizer (CT), $X^L_{\text{conv}} \in \mathcal{R}^{121\times 64}$, $W_{aL} \in \mathcal{R}^{64\times 1}$, and $W_{bL} \in \mathcal{R}^{64\times 64}$. In both cases, $X^L_{W_a} \in \mathcal{R}^{1\times 121}$ and $X^L_{W_b} \in \mathcal{R}^{121\times 64}$, and $X^L_{\text{cls}} \in \mathcal{R}^{1\times 64}$.

The `CLS` token, i.e., the LiDAR feature embeddings are then concatenated to the n HSI patch embeddings, making a total of $(n+1)$ patches from n, as shown in (5). A size of 64 is used, because it is a power of 2, which helps in calculating the head dimension

$$\widehat{X}^{HL} = \left[X^L_{\text{cls}} \parallel X^H_{\text{patch}}\right]. \tag{5}$$

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To retain the positional information, trainable position embeddings are added to the patch embeddings to preserve the semantic textural information of the image cube within the image patch tokens. For visual understanding, we refer to Fig. 1, where the position embeddings are added in elementwise fashion to all the patches (1 to $n + 1$). This process is followed by a dropout layer, which helps in reducing the effects of the overfitting of the network. The above process can be summarized by the following equation:

$$X^{HL} = \mathcal{DP}\left(X^{L}_{\text{cls}} \oplus \mathcal{PE} \;\|\; X^{H}_{\text{patch}} \oplus \mathcal{PE}\right) = \left(\widehat{X}^{L}_{\text{cls}} \;\|\; \widehat{X}^{H}_{\text{patch}}\right) \tag{6}$$

where $\mathcal{DP}$ is a dropout layer with a value of 0.1 and $\mathcal{PE}$ is a learnable position embedding.

Algorithm 1 Transformer Encoder With `mCrossPA`

Input: $\mathbf{X^{HL}}$, $\mathbf{X^{L}_{cls}}$, $\mathbf{X^{HL}_{patch}}$, h_d
Output: $\mathbf{y_{cls}}$
Multihead Cross Patch Attention (Begin)
Step 1.
$\mathbf{Q} = \mathbf{X^{L}_{cls}W_q}$, $\mathbf{K} = \mathbf{X^{HL}W_k}$, $\mathbf{V} = \mathbf{X^{HL}W_v}$,
Step 2. $\mathbf{Z} = \texttt{softmax}(\mathbf{QK^T}/\sqrt{\mathbf{h_d}})$, (7a)
$\texttt{mCrossPA}(\mathbf{X^{HL}}) = \mathcal{DP}(\mathbf{W_l ZV})$ (Eq. (7b))
Multihead Cross Patch Attention (End)
Step 3. $\mathbf{y'_{cls}} = \texttt{mCrossPA}(\mathbf{X^{HL}}))$ (Eq. (8a))
Step 4. $\mathbf{y_k} = f(y'_{cls}) \oplus ([\mathbf{\widehat{X}^{L}_{cls}} \;\|\; \mathbf{\widehat{X}^{H}_{patch}}])$ (Eq. (8b)),
$\mathbf{X^{HL}_k} = \mathbf{y_k} + \texttt{MLP}(\texttt{LN}(\mathbf{y_k}))$ (Eq. (9)),
$\mathbf{y_{cls}} = \mathbf{X^{HL}_k}(1, :)$

3) LiDAR and HSI CrossPA: The `CLS` (X^{L}_{cls}) token plays a vital role in learning the abstract representation of the entire HSI patch by exchanging the information between the patch tokens and itself. This entire process takes place in the transformer encoder blocks, where each transformer encoder block comprises a residual `mCrossPA` block and a residual `MLP` block, respectively. The residual attention block starts with a layer normalization (LN) operation followed by a self-attention layer, after which the output is added in elementwise ($\oplus$) fashion to the input of the LN (shown in Fig. 1). Similar to the attention block, the `MLP` block also starts with an LN operation, but the input of the LN is added elementwise ($\oplus$) to the output of `MLP` layer, as shown in Fig. 1.

Fig. 3 illustrates the CrossPA module for the LiDAR data (as a `CLS`) and for the HSI data (as a patch token). More specifically, the generated patch tokens from the HSI feature maps are collected and then concatenated with the external `CLS` tokens from the LiDAR data from the same spatial region corresponding to the HSI patch, followed by the combination of position embeddings and dropout, as shown in (5) and (6) and illustrated in Fig. 1. After that, the module performs `CrossPA` between X^{L}_{cls} and X^{HL}_{patch}, where query ($\mathbf{Q} = \mathbf{X^{L}_{cls}W}_q$ where $\mathrm{W_q} \in \mathcal{R}^{64\times64}$) along with the key ($\mathbf{K} = \mathbf{X}^{HL}\mathbf{W}_k$ where $\mathrm{W_k} \in \mathcal{R}^{64\times64}$) and the value ($\mathbf{V} = \mathbf{X}^{HL}\mathbf{W}_v$ where $\mathrm{W_v} \in \mathcal{R}^{64\times64}$) are fused to generate the `CLS` token (y_{cls}) at the end of the `CrossPA` block. A linear projection layer ($\mathbf{W}_l \in \mathcal{R}^{64\times64}$) is applied to the final fused

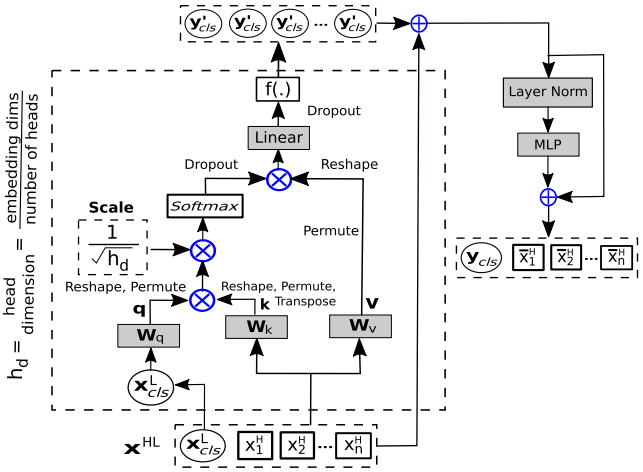


Fig. 3. `mCrossPA` taking example of LiDAR and HSI data.

`CLS` token, which precedes a dropout layer ($\mathcal{DP}$) with a value of 0.1. Mathematically, the cross attention can be formulated as follows:

$$Z = \text{softmax}\left(\frac{\mathbf{QK}^{\text{T}}}{\sqrt{\text{h}_\text{d}}}\right) \tag{7a}$$

$$\text{CrossPA}\left(\text{X}^{\text{HL}}\right) = \mathcal{DP}(\mathbf{W}_l\mathbf{ZV}) \tag{7b}$$

where $Z \in \mathcal{R}^{1\times64}$ and h_d = embedding dimension/number of heads. As shown in the self-attention module, if the number of heads is more than one, then the `CrossPA` will become a multihead cross attention, and, upon doing so, it can be represented as `mCrossPA`. The connections between the various patch tokens and CLS token are further strengthened by applying CrossPA with numerous heads. The output y_k of the `mCrossPA` module for a given embedding X^{HL}_{k-1} with LN and residual shortcut, where k is the kth transformer encoder block, is defined as follows:

$$y'_{\text{cls}} = \texttt{mCrossPA}\left(\text{LN}\left(X^{HL}_{k-1}\right)\right) \tag{8a}$$

$$y_k = f\left(y'_{\text{cls}}\right) \oplus \left[\widehat{X}^{L}_{\text{cls}} \;\|\; \widehat{X}^{H}_{\text{patch}}\right] \tag{8b}$$

where $y_k \in \mathcal{R}^{64\times64}$ and $f(.)$ is a function for broadcasting the dimension of y'_{cls} that is used to make the dimension the same as that of input X^{HL}_{k-1}, as shown in Fig. 3. This output y_k is then passed through multilayer perceptron (`MLP`) block after LN, followed by a residual shortcut to generate X^{HL}_{k}, which can be treated as input to $(k + 1)$th transformer encoder block, as shown in Fig. 3, and can be defined as follows:

$$X^{HL}_{k} = y_k \oplus \text{MLP}(LN(y_k)). \tag{9}$$

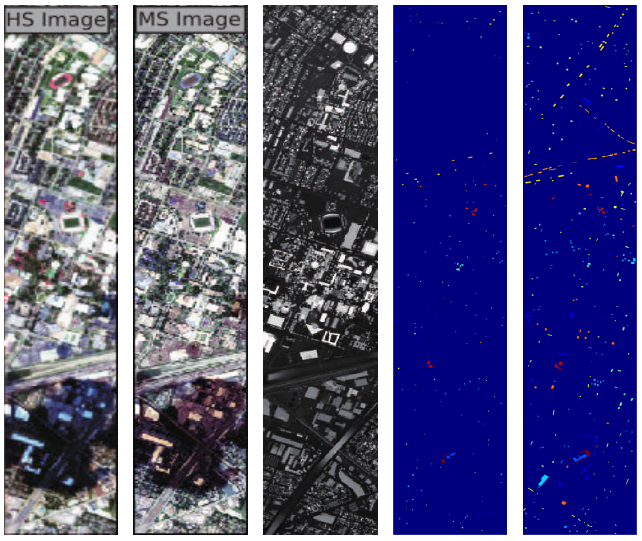
The number of heads h_d is set to 8 in the proposed model. Finally, the output from the kth transformer encoder blocks $y_{\text{cls}} = X^{HL}_{k}(1, :)$ is fed into the classifier head to get the final classification results.

IV. Experiments

A. HSI Datasets

In this section, we consider four different HSIs and their associated multimodal sources of data (LiDAR, MS, SAR,

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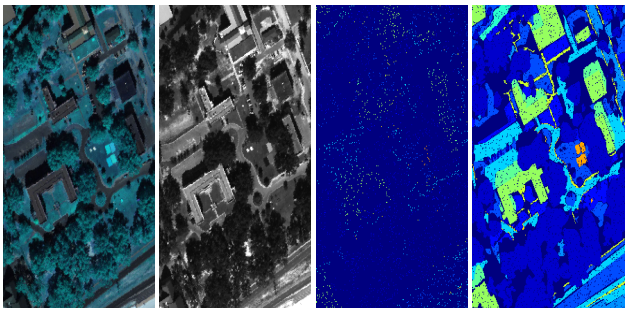
Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	662013	652648		Grass-healthy	198	1053
	Grass-stressed	190	1064		Grass-synthetic	192	505
	Tree	188	1056		Soil	186	1056
	Water	182	143		Residential	196	1072
	Commercial	191	1053		Road	193	1059
	Highway	191	1036		Railway	181	1054
	Parking-lot1	192	1041		Parking-lot2	184	285
	Tennis-court	181	247		Running-track	187	473

Fig. 4. UH scene. (a) Pseudo-color image from the HSI data using bands 64, 43, and 22, respectively. (b) Grayscale image from the MSI data. (c) Grayscale image from the LiDAR data. (d) Ground truth of disjoint training samples. (e) Ground truth of disjoint test samples. The table represents class-specific land-cover types and the number of disjoint training and test samples.

and DSM) to evaluate the performance of the proposed MFT network. The experimental datasets include the UH, Trento, MUUFL Gulfport, and Augsburg scenes.

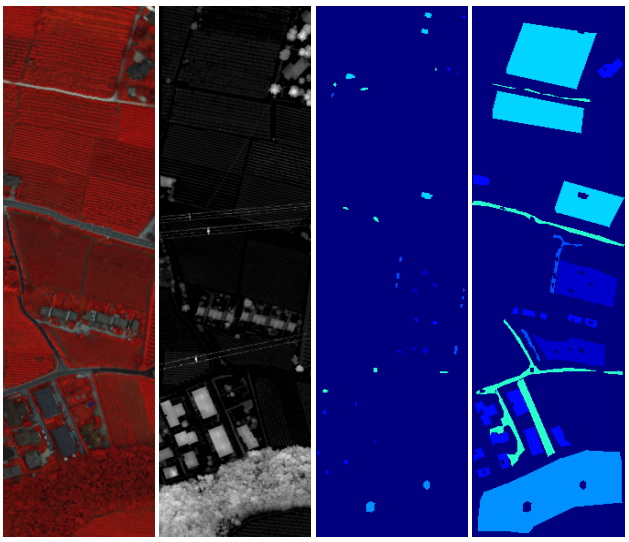
1) The UH dataset was collected by the Compact Airborne Spectrographic Imager (CASI) and was provided by the IEEE Geoscience and RS Society in 2013 as part of its Data Fusion Contest. The dataset comprises an HSI, an MS image, and a LiDAR image. All the images are made up of 340 × 1905 pixels, and the HSI has 144 bands, whereas the MS image has eight spectral bands. This dataset has a spatial resolution of 2.5 m per pixel and wavelength ranging between 0.38 and 1.05 μm. The ground truth has 15 different land-cover and land-use classes. Furthermore, the samples of the 15 land-cover classes are divided into fixed size training and testing samples. Fig. 4 represents the 15 different categories of land-cover and land-use and the training and test samples associated with each.

2) The MUUFL Gulfport scene was collected over the campus of the University of Southern Mississippi in November 2010 using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor [60], [61]. There are 325 × 220 pixels with 72 spectral bands in the HSI of this dataset. The LiDAR image of this dataset contains elevation data of two rasters. The eight initial and final bands were removed due to noise, giving a total of 64 bands. The data depicts 11 urban land-cover classes containing 53 687 ground-truth pixels. Fig. 5 displays the distribution of 5% of the samples randomly selected from every class.



Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	68817	20496		Trees	1162	22084
	Grass-Pure	214	4056		Grass-Groundsurface	344	6538
	Dirt-And-Sand	91	1735		Road-Materials	334	6353
	Water	23	443		Buildings'-Shadow	112	2121
	Buildings	312	5928		Sidewalk	69	1316
	Yellow-Curb	9	174		ClothPanels	13	256

Fig. 5. MUUFL scene. (a) True-color image from the HSI data using bands 40, 20, and 10, respectively. (b) Grayscale image from the LiDAR data. (c) Ground truth of MUUFL scene. (d) Test Samples. The table represents class-specific land-cover types and the number of randomly selected 5% training and the remaining 95% test samples.



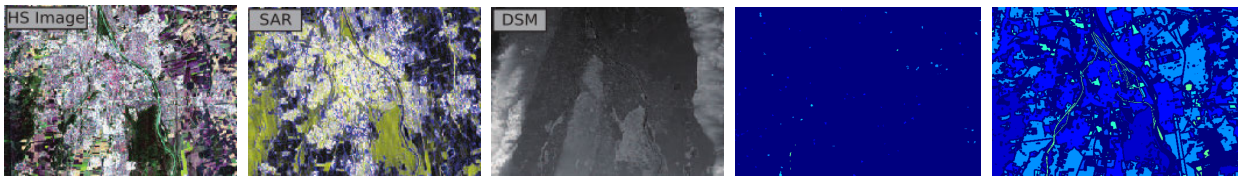
Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	98781	70205		Apples	129	3905
	Buildings	125	2778		Ground	105	374
	Woods	154	8969		Vineyard	184	10317
	Roads	122	3052				

Fig. 6. Trento data. (a) True-color image from the HSI using bands 40, 20, and 10, respectively. (b) Grayscale image from the LiDAR data. (c) Ground truth of disjoint training samples. (d) Ground truth of disjoint test samples. The table represents class-specific land-cover types and the number of disjoint training and test samples.

3) AISA Eagle sensors were used to collect HSI data over rural regions in the south of Trento, Italy, where the Optech ALTM 3100EA sensors collected LiDAR data. There are 63 bands in each HSI with wavelength ranging from 0.42 to 0.99 μm, and one raster in the LiDAR data that provides elevation information. The spectral resolution is 9.2 nm, and the spatial resolution is 1 m per pixel. The scene comprises six vegetation land-cover classes that are mutually exclusive and a pixel count of 600 × 166. Moreover, the training and test samples are disjoint. Fig. 6 provides information about the training and testing samples for each class.

4) There are three types of data in Augsburg scene, which include an HSI, a dual-polarimetric SAR (PolSAR) image,

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(a) (b) (c) (d) (e)

Color	Land cover	Train	Test	Color	Land cover	Train	Test
	Background	68816	20497		Forest	1162	22084
	Residential-Area	344	6538		Industrial-Area	91	1735
	Low-Plants	334	6353		Allotment	23	443
	Commercial-Area	112	2121		Water	312	5928

Fig. 7. Visualization of the Augsburg scene. (a) True-color image for the HS image data over bands 40, 20, and 10, respectively. (b) Grayscale image for the LiDAR data. (c) Grayscale image for the DSM data. (d) Training Image. (e) Test Image. The table represents class-specific land-cover types and the number of randomly selected 5% training and remaining 95% test samples.

and a DSM image [62]. SAR data are collected from the Sentinel-1 platform, while HS and DSM data are captured by DAS-EOC, German Aerospace Center (DLR), over the city of Augsburg, Germany. The collection is done by the HySpex sensor [63], the Sentinel-1 sensor, and the DLR-3 K system [64], respectively. The spatial resolutions of all images are downsampled to a unified spatial resolution of 30-m ground sampling distance (GSD) for adequately managing the multimodal fusion. For the HSI, there are 332×485 pixels and 180 spectral bands ranging between 0.4 and 2.5 μm. The DSM image has a single band, whereas the SAR image has four bands. The four bands indicate vertical-vertical (VV) intensity, vertical-horizontal (VH) intensity, the real component, and the imaginary component of the PolSAR covariance matrix's off-diagonal element. There are 15 distinct land-cover classes in the ground truth. Fig. 7 shows detailed information on the train and test sets.

A class-specific summary of the UH, MUUFL, and Trento scenes is shown in Figs. 4–7, respectively. Each dataset includes its corresponding ground truth, the type associated with the land-cover classes, and the number of available labeled samples per class.

B. Experimental Setup

In order to study the effectiveness of the proposed MFT model, we have performed extensive experiments and compared it with traditional as well as state-of-the-art methods. The compared methods include traditional classifiers, i.e., k-nearest neighbors (KNN) [65], RF [1], and SVM [27], as well as classical CNN models, i.e., CNN1D [33], CNN2D [34], CNN3D [35], and RNN [66]. We also considered a few state-of-the-art transformer models, i.e., ViT [49] and SpectralFormer [22]. The experiments have been performed using both HSI data and other sources of multimodal data, i.e., LiDAR, MSI, SAR, and DSM, combined. It can be noted that, for other compared models, the HSI and the other complementary data were concatenated bandwise before feeding the data to the models.

1) Configuration: All the tests have been performed on a Red Hat Enterprise Server (Release 7.6) with a CPU having ppc64le architecture and a total of 40 cores with four threads per core and 377 GB of RAM. The GPU used is a single Nvidia Tesla V100 with 32 510 MB of VRAM. The number of HSI patch tokens (n) that is received from the tokenization operation is set to 4 in our experiments.

A batch size of 64 and 500 has been used for training and testing the performance of the considered models, where patches of size $11 \times 11 \times B$ are extracted from the HSI and $11 \times 11 \times C$ from other sources of multimodal data. All the models except KNN, RF, SVM, and RNN have been trained with Adam optimizer [67], [68] with learning rate set to $5e^{-4}$ and weight decay of $5e^{-3}$. For the RNN, no weight decay was used, and a higher learning rate of $1e^{-3}$ was adopted. These models (including the RNN) also utilized a step scheduler with step size $=$ 50 and gamma $=$ 0.9, while training has been conducted using 500 epochs. Each experiment has been repeated three times, and the average and standard deviations are reported. The source code of the proposed MFT was implemented using PyTorch 1.5.0 and Python 3.7.7. The parameters and computational complexities for the considered models with respect to the dataset are discussed in Section IV-F.

2) Evaluation Matrix: Several widely used quantitative measures, such as overall accuracy (OA), average accuracy (AA), and statistical Kappa (κ) coefficients, are used to assess the performance of the proposed network and compare it with other methods. OA represents the proportion of correctly classified test samples versus all test samples, whereas AA represents the average of classwise accuracy. The kappa value reflects the degree of agreement between the generated classification maps of the considered model and the provided ground truth.

The experimental process has been conducted in three different settings: 1) disjointed (spatially and spectrally) training and test samples, i.e., the intersection between the training samples and testing samples remains empty; 2) different percentages of training samples have been randomly selected and used to validate the performance of the proposed network; and 3) different variations of the proposed model have been compared on the same disjointed datasets.

C. Performance Evaluation With Disjoint Train/Test Samples

Tables I–III report the quantitative OAs, AAs, κ values, and each class accuracy using the proposed model as well as the other methods for the HSI, the HSI with LiDAR data, and HSI

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TABLE I

OA, AA, AND KAPPA VALUES ON THE UH DATASET (IN %) BY CONSIDERING HSI DATA ONLY

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	77.87	82.81 ± 00.08	79.77	81.10 ± 00.43	80.53 ± 00.23	81.70 ± 00.38	80.22 ± 02.80	82.40 ± 00.39	82.49 ± 00.25	**82.53 ± 00.13**
2	77.44	82.86 ± 00.36	82.42	80.23 ± 01.51	83.90 ± 00.09	80.55 ± 01.84	78.51 ± 00.19	80.29 ± 00.79	**89.13 ± 06.36**	85.03 ± 00.04
3	96.83	63.10 ± 01.71	59.41	53.73 ± 00.09	57.49 ± 02.10	96.57 ± 00.09	52.94 ± 17.72	97.43 ± 01.32	69.77 ± 12.29	**98.55 ± 01.08**
4	75.28	91.95 ± 00.15	83.81	83.74 ± 00.71	89.46 ± 00.24	78.54 ± 02.82	83.81 ± 07.71	90.40 ± 00.25	91.73 ± 03.42	**94.60 ± 02.53**
5	90.72	99.78 ± 00.12	95.27	87.06 ± 01.51	92.36 ± 01.20	98.48 ± 00.28	85.61 ± 09.73	99.24 ± 00.13	96.78 ± 00.76	**99.84 ± 00.16**
6	66.43	96.97 ± 00.33	67.13	52.45 ± 01.14	64.10 ± 02.38	73.89 ± 00.66	70.16 ± 07.18	91.38 ± 00.87	85.31 ± 07.99	**95.10 ± 00.57**
7	76.96	85.23 ± 00.50	83.21	71.42 ± 01.46	71.39 ± 04.04	82.77 ± 01.45	73.01 ± 01.31	**86.10 ± 01.54**	80.25 ± 02.23	85.79 ± 00.94
8	30.96	42.58 ± 00.36	29.53	41.12 ± 01.70	44.95 ± 06.09	38.30 ± 01.09	43.84 ± 08.75	73.95 ± 00.29	62.74 ± 11.68	**79.87 ± 09.29**
9	69.50	85.36 ± 00.13	75.45	60.25 ± 00.08	62.45 ± 02.04	65.94 ± 02.86	68.84 ± 02.54	85.33 ± 02.24	70.57 ± 01.62	**89.05 ± 02.47**
10	42.95	35.81 ± 00.70	46.62	39.12 ± 02.03	49.94 ± 01.66	43.28 ± 07.46	37.52 ± 01.15	50.42 ± 05.59	48.17 ± 05.04	**62.45 ± 02.95**
11	56.17	63.03 ± 00.39	45.07	42.06 ± 00.78	44.53 ± 00.78	33.59 ± 02.72	49.65 ± 09.61	80.80 ± 03.06	62.75 ± 04.12	**98.04 ± 01.45**
12	75.79	66.63 ± 00.50	70.03	62.98 ± 04.58	53.92 ± 06.25	67.85 ± 02.10	64.07 ± 00.28	81.91 ± 01.77	79.09 ± 02.16	**95.84 ± 00.48**
13	60.35	87.60 ± 00.17	68.42	42.11 ± 01.52	47.13 ± 02.33	77.54 ± 01.98	53.92 ± 08.72	89.47 ± 01.74	63.63 ± 05.56	**91.81 ± 00.33**
14	76.92	99.73 ± 00.19	75.30	83.94 ± 00.38	82.46 ± 03.07	92.58 ± 01.06	81.38 ± 12.63	99.33 ± 00.95	93.66 ± 04.30	**99.73 ± 00.19**
15	88.37	85.62 ± 01.38	49.89	34.46 ± 02.04	42.92 ± 05.11	93.52 ± 01.55	44.12 ± 13.23	**99.72 ± 00.26**	77.24 ± 07.34	92.60 ± 02.42
OA	69.48	74.87 ± 00.09	68.13	63.04 ± 00.08	65.85 ± 00.09	70.26 ± 00.14	65.20 ± 02.74	83.23 ± 00.47	76.35 ± 01.97	**88.45 ± 00.28**
AA	70.84	77.94 ± 00.14	67.42	61.05 ± 00.10	64.50 ± 00.11	73.67 ± 00.17	64.51 ± 02.77	85.88 ± 00.33	76.89 ± 02.31	**90.05 ± 00.29**
$\kappa(\times 100)$	67.08	72.93 ± 00.09	65.56	60.01 ± 00.08	63.04 ± 00.09	67.91 ± 00.16	62.43 ± 02.87	81.88 ± 00.51	74.42 ± 02.14	**87.46 ± 00.30**

TABLE II

OA, AA, AND KAPPA VALUES ON THE UH DATASET (IN %) BY CONSIDERING HSI AND LIDAR DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	77.30	79.33 ± 00.38	79.96	81.32 ± 00.16	82.62 ± 00.08	82.30 ± 00.29	81.80 ± 00.62	82.59 ± 00.18	**82.65 ± 00.24**	82.34 ± 00.47
2	81.58	71.49 ± 01.69	82.89	81.92 ± 00.19	81.83 ± 00.66	78.70 ± 00.98	71.40 ± 05.21	82.33 ± 01.46	83.33 ± 00.29	**88.78 ± 05.27**
3	97.82	98.15 ± 00.25	60.79	57.49 ± 00.61	62.90 ± 00.37	96.96 ± 00.47	76.04 ± 14.49	97.43 ± 00.70	75.78 ± 12.48	**98.15 ± 01.08**
4	80.59	78.09 ± 00.47	86.65	86.74 ± 00.23	88.57 ± 00.04	80.49 ± 01.57	88.51 ± 02.14	92.93 ± 01.72	91.10 ± 01.29	**94.35 ± 02.40**
5	91.86	89.68 ± 00.08	95.36	95.93 ± 00.27	97.19 ± 00.19	98.11 ± 00.54	85.76 ± 04.87	**99.84 ± 00.04**	98.30 ± 01.12	99.12 ± 01.25
6	78.32	93.71 ± 00.57	69.23	58.04 ± 00.57	65.03 ± 01.98	73.89 ± 01.84	85.78 ± 02.87	84.15 ± 03.30	89.04 ± 02.70	**99.30 ± 00.00**
7	80.97	79.32 ± 01.03	85.26	77.95 ± 00.88	80.22 ± 01.42	81.09 ± 00.75	82.77 ± 01.71	87.84 ± 01.49	81.72 ± 01.19	**88.56 ± 01.16**
8	42.45	53.37 ± 02.26	49.38	54.42 ± 02.33	60.81 ± 04.32	44.63 ± 06.52	61.44 ± 07.10	79.93 ± 00.16	67.81 ± 09.00	**86.89 ± 05.33**
9	69.22	76.27 ± 00.46	78.28	66.13 ± 00.19	67.74 ± 01.08	74.76 ± 03.84	67.42 ± 07.89	82.94 ± 00.85	74.47 ± 01.89	**87.91 ± 03.90**
10	43.44	38.42 ± 01.07	50.48	47.30 ± 02.18	51.74 ± 01.33	37.52 ± 11.23	38.45 ± 03.12	52.93 ± 05.14	56.76 ± 06.31	**64.70 ± 00.92**
11	65.84	65.02 ± 00.47	49.34	44.40 ± 02.06	39.91 ± 03.52	40.80 ± 08.50	64.39 ± 06.38	80.99 ± 03.06	59.93 ± 07.67	**98.64 ± 00.70**
12	81.94	82.20 ± 00.56	72.05	63.66 ± 08.15	82.20 ± 03.85	66.38 ± 02.36	77.07 ± 05.79	91.07 ± 02.55	70.00 ± 02.98	**94.24 ± 02.19**
13	60.35	67.72 ± 00.50	77.19	50.41 ± 01.86	52.40 ± 01.29	68.77 ± 12.16	47.13 ± 03.73	87.84 ± 01.91	66.20 ± 00.44	**90.29 ± 00.66**
14	83.81	82.46 ± 00.83	75.30	86.50 ± 00.69	92.44 ± 02.89	92.85 ± 01.16	97.98 ± 00.87	**100.0 ± 00.00**	92.04 ± 04.50	99.73 ± 00.38
15	91.33	95.56 ± 00.30	55.18	40.10 ± 03.20	52.08 ± 04.14	96.90 ± 01.41	73.50 ± 09.82	**99.65 ± 00.50**	77.45 ± 13.31	99.58 ± 00.17
OA	73.50	73.83 ± 00.31	71.94	68.11 ± 00.29	71.88 ± 00.36	71.41 ± 00.35	72.31 ± 02.44	85.05 ± 00.47	76.87 ± 02.21	**89.80 ± 00.53**
AA	75.12	76.72 ± 00.28	71.16	66.15 ± 00.09	70.51 ± 00.27	74.28 ± 00.30	73.30 ± 02.50	86.83 ± 00.53	77.77 ± 02.78	**91.51 ± 00.40**
$\kappa(\times 100)$	71.41	71.78 ± 00.33	69.67	65.48 ± 00.31	69.56 ± 00.39	69.12 ± 00.36	70.14 ± 02.62	83.84 ± 00.51	75.03 ± 02.37	**88.93 ± 00.59**

TABLE III

OA, AA, AND KAPPA VALUES ON THE UH DATASET (IN %) BY CONSIDERING HSI AND MS IMAGE DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	78.06	78.82 ± 00.40	79.87	84.52 ± 02.26	88.86 ± 04.90	82.15 ± 00.68	79.23 ± 00.39	**89.02 ± 05.03**	81.54 ± 00.65	82.72 ± 00.41
2	75.28	72.65 ± 02.53	79.14	81.05 ± 02.70	**86.18 ± 05.21**	84.62 ± 00.04	81.42 ± 00.51	82.11 ± 00.67	80.20 ± 02.86	85.09 ± 00.09
3	96.63	96.63 ± 00.49	61.25	38.75 ± 04.96	45.35 ± 03.93	49.70 ± 04.53	38.75 ± 01.22	**99.21 ± 00.16**	59.27 ± 11.24	98.55 ± 00.76
4	68.47	77.15 ± 00.27	85.23	81.09 ± 03.63	75.47 ± 06.68	86.68 ± 00.50	88.35 ± 01.54	81.72 ± 01.08	90.59 ± 00.31	**95.99 ± 02.39**
5	89.77	90.03 ± 00.89	87.22	72.92 ± 04.99	87.12 ± 04.46	92.20 ± 03.75	90.91 ± 01.01	93.15 ± 00.97	95.01 ± 00.96	**99.78 ± 00.09**
6	71.33	86.48 ± 06.49	61.54	63.64 ± 02.97	72.03 ± 04.67	79.95 ± 02.87	77.39 ± 02.87	**99.77 ± 00.33**	73.43 ± 02.06	97.20 ± 01.98
7	72.57	77.89 ± 03.73	81.53	58.18 ± 01.21	68.13 ± 05.26	78.17 ± 04.71	65.76 ± 03.53	72.45 ± 05.50	78.42 ± 00.62	**86.32 ± 00.69**
8	30.10	41.63 ± 00.25	18.80	26.34 ± 16.50	23.36 ± 15.68	27.95 ± 06.36	35.61 ± 01.58	77.87 ± 02.72	48.40 ± 01.57	**81.16 ± 08.04**
9	68.18	74.66 ± 00.40	68.27	54.93 ± 06.23	59.05 ± 04.45	83.41 ± 00.76	72.05 ± 03.16	83.47 ± 07.68	73.25 ± 01.69	**87.76 ± 00.74**
10	42.28	38.42 ± 00.21	48.46	28.73 ± 04.69	32.37 ± 01.22	69.76 ± 04.74	34.04 ± 05.16	51.03 ± 01.04	50.87 ± 01.76	**74.71 ± 15.98**
11	53.80	58.16 ± 00.54	38.71	42.50 ± 01.88	43.33 ± 01.90	60.25 ± 04.92	36.78 ± 02.50	76.03 ± 01.53	55.03 ± 00.43	**93.71 ± 06.01**
12	68.68	80.24 ± 00.09	57.54	43.84 ± 13.82	35.57 ± 11.98	65.10 ± 15.37	58.44 ± 03.47	75.47 ± 02.24	71.41 ± 03.96	**96.16 ± 00.87**
13	57.89	66.08 ± 01.47	65.96	48.65 ± 04.63	52.28 ± 04.22	47.49 ± 10.75	52.63 ± 06.82	83.63 ± 04.12	57.08 ± 00.17	**92.51 ± 00.17**
14	79.35	80.30 ± 01.34	83.40	63.16 ± 02.17	70.58 ± 12.48	96.22 ± 03.34	81.24 ± 07.20	99.60 ± 00.33	88.26 ± 08.02	**100.0 ± 00.00**
15	86.68	**94.15 ± 00.20**	48.41	55.60 ± 02.85	45.24 ± 09.15	58.42 ± 08.47	40.94 ± 15.84	91.97 ± 01.70	46.51 ± 05.61	86.82 ± 08.99
OA	67.16	71.51 ± 00.46	64.18	56.63 ± 00.61	59.06 ± 00.57	71.49 ± 01.07	62.61 ± 01.80	80.49 ± 01.19	70.95 ± 00.58	**89.15 ± 00.96**
AA	69.27	74.22 ± 00.06	64.36	56.26 ± 00.93	58.99 ± 00.25	70.80 ± 01.16	62.24 ± 02.51	83.77 ± 01.15	69.95 ± 00.63	**90.56 ± 00.93**
$\kappa(\times 100)$	64.61	69.30 ± 00.42	61.30	53.07 ± 00.68	55.70 ± 00.60	69.02 ± 01.16	59.64 ± 01.98	78.88 ± 01.29	68.59 ± 00.64	**88.22 ± 01.04**

with MS data, respectively, for the disjoint Houston dataset. The best results are shown in bold. The evaluation data indicate that the proposed method outperforms the rest by obtaining the highest OAs, AAs, and κ values and outperforms others in most classwise classification accuracies.

Overall, the transformer networks outperform the conventional networks and classifiers. The proposed model dominates the classwise accuracies and surpasses the other methods in terms of OA, AA, and Kappa in all three cases for the Houston dataset, i.e., HSI, HSI with LiDAR, and HSI with MS

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TABLE IV
OA, AA, AND KAPPA VALUES ON THE UT DATASET (IN %) BY CONSIDERING HS IMAGE ONLY

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	89.07	96.65 ± 00.36	97.21	97.29 ± 00.20	96.76 ± 00.24	92.22 ± 00.66	88.06 ± 04.36	89.07 ± 00.86	94.01 ± 03.77	**98.20 ± 00.64**
2	88.59	89.38 ± 00.74	94.17	86.73 ± 01.60	84.17 ± 01.05	91.42 ± 00.69	79.11 ± 06.15	84.74 ± 06.22	91.26 ± 01.23	**91.65 ± 01.43**
3	87.17	73.71 ± 01.61	56.15	50.62 ± 00.13	50.53 ± 02.84	**96.88 ± 00.91**	39.22 ± 19.36	92.25 ± 01.65	46.52 ± 05.46	94.92 ± 00.95
4	80.24	99.92 ± 00.02	84.78	99.17 ± 00.14	96.12 ± 00.24	99.62 ± 00.10	83.31 ± 04.10	**99.63 ± 00.21**	85.48 ± 11.89	99.45 ± 00.25
5	95.15	99.96 ± 00.01	98.13	99.18 ± 00.02	98.76 ± 00.16	98.99 ± 00.12	97.96 ± 00.64	98.23 ± 00.10	97.60 ± 01.45	**99.72 ± 00.10**
6	69.59	66.73 ± 01.52	55.05	59.62 ± 01.47	66.19 ± 00.90	85.47 ± 00.62	70.87 ± 15.09	84.04 ± 06.23	61.42 ± 01.79	**91.52 ± 01.72**
OA	86.42	94.73 ± 00.14	88.55	93.02 ± 00.06	92.31 ± 00.05	96.14 ± 00.02	86.83 ± 01.91	94.62 ± 00.21	88.42 ± 03.45	**97.76 ± 00.40**
AA	84.97	87.73 ± 00.43	80.91	82.10 ± 00.08	82.09 ± 00.46	94.10 ± 00.14	76.42 ± 04.46	91.33 ± 00.22	79.38 ± 00.96	**95.91 ± 00.41**
$\kappa(\times 100)$	82.21	92.92 ± 00.18	84.83	90.65 ± 00.09	89.69 ± 00.07	94.83 ± 00.02	82.49 ± 02.53	92.81 ± 00.28	84.68 ± 04.41	**97.00 ± 00.53**

image. Combining HSI data with LiDAR data, even using the traditional fusion method, leads to performance gains in almost all models. Though the proposed model stands out on top in the case of Houston using HSI with LiDAR data, the increase in performance (1.35% OA, 1.46% AA, and 1.47% κ) is not significantly better than that achieved by the other models. The new transformer-based fusion technique used in the proposed model provides the best result for Houston using HSI data with MS image, where almost all the other models that use the traditional fusion method have lower OAs, AAs, and κ values when compared with those in the case Houston using HSI data only, but the gains obtained by the proposed model remain consistent.

It can be seen from Table I that, among the conventional classifiers, RF performs the best, having the mean OA, AA, and κ of 74.87%, 77.94%, and 72.93% with the standard deviations of 0.09%, 0.14%, and 0.09%, respectively. RF can also beat all the classical networks shown in the table but is not as good when compared with transformer networks. The proposed transformer network provides the best performance among the three transformer-based networks giving the mean OA, AA, and κ of 88.45%, 90.05%, and 87.46% with the standard deviations of 0.28%, 0.14%, and 0.09%, respectively, followed by ViT and SpectralFormer. The standard deviations of the proposed model are also lower than those of the other transformer networks, since our model can represent long-range dependencies between the feature maps better than ViT and SpectralFormer. Conventional classifiers, which rely completely on the spectral representation of HSI data, cannot capture the spatial information and, hence, perform worse than transformer networks.

Table II indicates a substantial rise in the accuracies of all the given classical networks, where the accuracies of the best-performing classical networks are close to those of the best-performing conventional classifiers. Table I confirms that RF outperforms the conventional classifiers as well as the classical networks due to its better discriminative feature learning, but still cannot compete with the transformer networks. The proposed model again beats the rest by quite a margin due to its superior learning of spectral and spatial information, with the mean OA, AA, and κ of 89.80%, 91.51%, and 88.93% with the minimal standard deviations of 0.53%, 0.40%, and 0.59%, respectively.

Table III reveals an opposite trend than Table II in all but two networks. The networks other than CNN3D and the proposed network perform worse with the fusion of both HSI and MS image using the traditional fusion method when compared with that of HSI only (Table I). Unsurprisingly, the novel transformer-based fusion technique allows the proposed model to increase its performance resulting in the mean OA, AA, and κ of 89.15%, 90.56%, and 88.22% with the acceptable standard deviations of 0.96%, 0.93%, and 1.04%, respectively.

To demonstrate the generalization ability of the proposed model for RS image classification methods, we have considered Trento and MUUFL datasets (HSI data only and HSI with LiDAR data) with disjoint training and test samples. Fig. 6 shows the pseudo-color map, corresponding LiDAR data, and the disjoint train and test ground truth for the Trento dataset. Fig. 6 also provides six different land-cover classes and the number of samples belonging to the train and test sets for each class in a table. Fig. 5 shows the pseudo-color map corresponding to the LiDAR image, and the disjoint train and test ground truth for the MUUFL dataset. It also gives the 11 different land-cover classes and the number of training and testing samples per class.

Tables IV and VI show the experimental results for Trento and MUUFL datasets using HS images only. Among conventional classifiers and classical networks, CNN3D performs the best for Trento. In the case of MUUFL, both RF and RNN perform similarly and better than the other conventional classifiers and classical networks. In the case of transformer-based models, the proposed model takes the lead with OA = 97.76% ± 00.40%, AA = 95.91% ± 00.41%, and κ = 97.00% ± 00.53% for Trento and OA = 94.18% ± 00.09%, AA = 81.00% ± 00.55%, and κ = 92.30% ± 00.12% for MUUFL.

Tables V and VII show the experimental results for Trento and MUUFL datasets using combined HS image and LiDAR data. In Trento, except for RF, it is quite evident from the better accuracies of all the models that using LiDAR and HS images improves all models' performance. But, even with this significant increase in performance, the proposed model (in Tables IV and VI) performs better than the rest of the models, while the proposed model (in Tables V and VII) performs even better. For the Trento data using HS image and LiDAR, the gain in the proposed model's OA is 0.56%, AA is 0.07%, and κ is 0.75%. For MUUFL using HS image and LiDAR data, the gain in the proposed model's OA, AA, and κ is 0.26%, 0.48%, and 0.21%, respectively. Among conventional classifiers and

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TABLE V

OA, AA, AND KAPPA VALUES ON THE UT DATASET (IN %) BY CONSIDERING HS IMAGE AND LIDAR DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	87.94	83.73 ± 00.06	97.44	97.00 ± 00.50	96.98 ± 00.21	92.95 ± 00.10	91.75 ± 04.30	90.87 ± 00.77	96.76 ± 01.71	**98.23 ± 00.38**
2	95.79	96.30 ± 00.06	98.12	96.51 ± 01.70	97.56 ± 00.14	98.09 ± 00.23	**99.47 ± 00.37**	99.32 ± 00.77	97.25 ± 00.66	99.34 ± 00.02
3	81.28	70.94 ± 01.55	56.15	42.34 ± 06.33	55.35 ± 00.00	**93.85 ± 01.09**	79.23 ± 16.47	92.69 ± 01.53	58.47 ± 11.54	89.84 ± 09.00
4	96.25	99.73 ± 00.07	97.53	99.77 ± 00.05	99.66 ± 00.03	99.32 ± 00.05	99.58 ± 00.42	**100.0 ± 00.00**	99.24 ± 00.21	99.82 ± 00.26
5	95.29	95.35 ± 00.25	98.13	99.27 ± 00.09	99.56 ± 00.07	98.74 ± 00.04	98.39 ± 00.65	97.77 ± 00.86	93.52 ± 01.75	**99.93 ± 00.05**
6	83.85	72.63 ± 00.90	78.96	76.91 ± 03.62	76.91 ± 00.15	88.15 ± 00.20	85.86 ± 02.89	86.72 ± 02.02	73.39 ± 06.78	**88.72 ± 00.94**
OA	93.29	92.57 ± 00.07	95.33	95.81 ± 00.13	96.14 ± 00.03	96.93 ± 00.03	96.43 ± 00.79	96.47 ± 00.49	93.51 ± 01.27	**98.32 ± 00.25**
AA	90.07	86.45 ± 00.32	87.72	85.30 ± 00.72	87.67 ± 00.04	95.18 ± 00.18	92.38 ± 03.50	94.56 ± 00.57	86.44 ± 02.96	**95.98 ± 01.64**
$\kappa(\times 100)$	91.11	90.11 ± 00.09	93.76	94.39 ± 00.17	94.83 ± 00.04	95.89 ± 00.04	95.21 ± 01.06	95.28 ± 00.65	91.36 ± 01.67	**97.75 ± 00.33**

TABLE VI

OA, AA, AND KAPPA VALUES ON THE MUUFL DATASET (IN %) BY CONSIDERING HS IMAGE DATA ONLY

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	92.74	97.97 ± 00.07	96.64	95.21 ± 00.05	95.94 ± 00.20	94.72 ± 00.65	96.06 ± 00.19	97.40 ± 00.25	94.63 ± 00.89	**97.61 ± 00.15**
2	47.46	77.86 ± 00.19	59.37	69.31 ± 01.06	70.87 ± 01.50	62.98 ± 03.76	80.34 ± 02.85	77.34 ± 01.62	77.12 ± 03.77	**92.51 ± 00.87**
3	69.10	84.33 ± 00.23	81.49	74.31 ± 00.60	78.65 ± 00.78	71.15 ± 03.46	80.65 ± 00.87	86.10 ± 01.49	74.26 ± 02.81	**92.12 ± 01.66**
4	53.43	86.09 ± 00.37	74.35	78.71 ± 01.27	82.96 ± 01.17	64.86 ± 02.15	86.09 ± 01.95	**93.35 ± 00.84**	84.59 ± 01.18	92.83 ± 01.30
5	83.68	92.10 ± 00.08	83.80	75.84 ± 01.93	78.76 ± 01.09	77.95 ± 02.15	90.15 ± 00.39	**94.82 ± 00.77**	89.38 ± 02.06	94.31 ± 01.26
6	01.13	68.77 ± 00.65	15.35	46.35 ± 00.74	48.53 ± 02.60	75.24 ± 90.92	53.42 ± 04.33	82.69 ± 01.31	60.42 ± 09.08	**88.56 ± 02.86**
7	43.33	80.47 ± 00.42	77.09	78.45 ± 00.58	78.96 ± 00.30	48.44 ± 01.87	80.39 ± 02.75	87.44 ± 00.97	83.77 ± 03.58	**92.68 ± 01.42**
8	73.48	91.57 ± 00.17	87.11	68.28 ± 02.36	68.86 ± 00.37	69.48 ± 01.91	88.67 ± 02.04	**97.41 ± 00.75**	88.91 ± 02.78	97.08 ± 00.87
9	04.94	45.87 ± 00.31	21.50	39.97 ± 00.27	45.64 ± 00.48	04.79 ± 03.39	58.94 ± 00.73	58.92 ± 07.76	55.02 ± 01.43	**59.80 ± 01.56**
10	00.00	04.98 ± 03.05	00.00	08.81 ± 00.72	13.03 ± 00.27	00.00 ± 00.00	22.80 ± 04.51	**32.76 ± 07.01**	17.62 ± 02.12	12.45 ± 04.79
11	58.59	48.31 ± 02.12	61.33	23.83 ± 03.24	24.74 ± 00.97	61.20 ± 02.17	**80.99 ± 04.01**	66.67 ± 04.94	45.44 ± 07.29	71.09 ± 01.66
OA	75.80	89.85 ± 00.03	84.30	81.17 ± 00.09	82.95 ± 00.12	77.59 ± 00.05	88.60 ± 00.69	91.99 ± 00.35	86.68 ± 00.93	**94.18 ± 00.09**
AA	47.99	70.76 ± 00.29	59.82	59.92 ± 00.52	62.45 ± 00.22	50.58 ± 00.34	74.41 ± 01.40	79.54 ± 01.93	70.11 ± 01.52	**81.00 ± 00.55**
$\kappa(\times 100)$	67.34	86.44 ± 00.04	78.89	74.95 ± 00.12	77.30 ± 00.16	69.81 ± 00.10	84.90 ± 00.92	89.37 ± 00.46	82.43 ± 01.19	**92.30 ± 00.12**

TABLE VII

OA, AA, AND KAPPA VALUES ON THE MUUFL DATASET (IN %) BY CONSIDERING HS IMAGE AND LIDAR DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	92.12	95.42 ± 00.09	96.63	95.05 ± 00.22	95.79 ± 00.11	95.10 ± 00.19	95.84 ± 00.14	97.85 ± 00.29	97.30 ± 00.83	**97.90 ± 00.39**
2	51.85	74.03 ± 00.11	59.25	70.35 ± 03.30	72.76 ± 00.58	63.72 ± 03.18	81.93 ± 02.11	76.06 ± 02.40	69.35 ± 05.16	**92.11 ± 01.58**
3	69.35	75.81 ± 00.38	81.46	75.80 ± 02.09	78.92 ± 00.52	69.94 ± 03.42	80.47 ± 02.13	87.58 ± 03.46	78.48 ± 03.41	**91.80 ± 00.82**
4	57.00	68.59 ± 00.77	73.54	78.60 ± 02.77	83.59 ± 00.99	63.90 ± 01.84	87.01 ± 01.46	**92.05 ± 02.31**	82.63 ± 03.68	91.59 ± 02.25
5	83.87	88.17 ± 00.18	83.79	78.31 ± 01.48	78.29 ± 01.12	79.48 ± 01.43	90.65 ± 00.65	94.73 ± 00.60	87.91 ± 02.97	**95.60 ± 01.21**
6	19.19	77.28 ± 00.93	15.35	46.35 ± 02.49	50.34 ± 02.13	02.86 ± 03.00	54.25 ± 03.14	82.02 ± 01.13	58.77 ± 02.76	**88.19 ± 03.49**
7	44.60	64.83 ± 00.97	77.04	78.31 ± 00.20	79.70 ± 00.26	47.96 ± 01.00	81.24 ± 01.32	87.11 ± 01.54	85.87 ± 00.62	**90.27 ± 02.13**
8	76.97	93.29 ± 00.27	86.94	66.72 ± 01.17	71.95 ± 01.10	70.47 ± 01.25	88.39 ± 01.50	**97.60 ± 00.16**	95.60 ± 01.26	97.26 ± 00.53
9	09.95	19.15 ± 01.37	21.28	40.15 ± 02.96	43.92 ± 01.24	06.28 ± 04.91	60.54 ± 04.40	57.83 ± 04.45	53.52 ± 04.32	**61.35 ± 03.80**
10	00.00	04.41 ± 00.72	00.00	09.20 ± 01.24	12.45 ± 00.27	00.00 ± 00.00	26.44 ± 02.82	**31.99 ± 08.86**	08.43 ± 02.22	17.43 ± 04.63
11	64.45	71.88 ± 00.84	62.89	25.65 ± 02.89	26.82 ± 02.60	66.93 ± 01.76	**87.50 ± 02.92**	58.72 ± 03.85	35.29 ± 06.00	72.79 ± 09.25
OA	76.83	85.32 ± 00.09	84.24	81.50 ± 00.03	83.40 ± 00.04	77.99 ± 00.06	88.79 ± 00.45	92.15 ± 00.19	88.25 ± 00.56	**94.34 ± 00.07**
AA	51.76	66.62 ± 00.16	59.83	60.41 ± 00.48	63.14 ± 00.21	51.51 ± 00.40	75.84 ± 00.62	78.50 ± 01.28	68.47 ± 01.44	**81.48 ± 00.70**
$\kappa(\times 100)$	68.92	80.39 ± 00.12	78.80	75.43 ± 00.07	77.94 ± 00.06	70.31 ± 00.03	85.18 ± 00.60	89.56 ± 00.27	84.40 ± 00.77	**92.51 ± 00.10**

classical networks, SVM and CNN3D perform the best for Trento, while MUUFL, RF, and CNN2D perform the best.

The proposed MFT is not only limited to HS image, MS image, and LiDAR data but can also work with SAR and DSM data. To illustrate this capability of the proposed MFT, we have considered the Augsburg dataset (using HS image data only, HS image and SAR data, and HS image and DSM data) with disjoint training and test samples. Fig. 7 shows the pseudo-color map, corresponding SAR data, corresponding DSM data, and the disjoint train and test ground truth for the Augsburg dataset. It also provides the seven different land-cover classes and the number of samples belonging to the train and test set per class.

After observing the AA in Table VIII, it is evident that Augsburg is a considerably harder dataset than the previous ones. Consequently, the gain in performance in most models with fusion is not significant. However, the proposed model gives the highest classwise classification accuracies for most classes in all three cases. It also provides the highest OAs, AAs, and κ values with the exception in the case of the Augsburg dataset using HS and SAR data. Table IX indicates that ViT has 0.76% more AA than the proposed model. However, the proposed MFT has 4.39% and 6.20% more OA and κ than ViT, respectively. In Table X, the AA of the proposed MFT reaches 64.70% ± 00.44%, which is the highest recorded for all three cases in all models.

RNN performs the worst in Tables VIII and X but gets a significant boost in performance in Table IX. Comparing Tables IX and X, conventional classifiers, such as KNN and SVM, show barely noticeable changes in performance. On the

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TABLE VIII

OA, AA, AND KAPPA VALUES ON THE AUGSBURG DATASET (IN %) BY CONSIDERING HS IMAGE DATA ONLY

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	83.27	90.18 ± 00.34	91.51	77.79 ± 04.77	82.77 ± 02.64	78.66 ± 01.26	21.21 ± 29.99	88.62 ± 03.36	83.23 ± 04.20	**93.93 ± 00.62**
2	86.38	97.41 ± 00.14	86.18	87.40 ± 01.55	86.19 ± 01.41	96.10 ± 00.98	**98.99 ± 01.39**	94.90 ± 00.84	86.63 ± 02.81	96.97 ± 00.49
3	36.79	04.59 ± 00.33	10.34	30.30 ± 08.46	56.08 ± 00.99	49.25 ± 23.59	00.00 ± 00.00	68.15 ± 03.93	27.75 ± 05.25	**68.34 ± 01.97**
4	49.72	76.16 ± 00.99	62.75	65.25 ± 01.80	64.95 ± 02.05	84.96 ± 01.88	05.35 ± 07.56	84.40 ± 02.26	60.96 ± 04.83	**93.14 ± 01.80**
5	38.62	18.16 ± 00.62	23.14	15.74 ± 04.72	10.26 ± 02.98	10.13 ± 09.16	00.00 ± 00.00	34.29 ± 02.60	37.54 ± 07.18	**46.02 ± 05.76**
6	08.30	00.12 ± 00.09	00.00	**20.35 ± 13.47**	14.12 ± 08.17	08.45 ± 03.66	00.00 ± 00.00	17.83 ± 05.12	03.24 ± 00.90	14.59 ± 04.29
7	05.37	08.03 ± 00.99	10.68	14.36 ± 02.30	24.62 ± 02.92	11.70 ± 04.50	00.00 ± 00.00	**45.90 ± 00.92**	13.49 ± 08.87	26.45 ± 15.31
OA	67.27	79.96 ± 00.23	71.60	72.00 ± 01.70	73.59 ± 00.59	82.89 ± 00.78	40.26 ± 02.03	85.90 ± 00.26	70.81 ± 01.47	**90.26 ± 00.91**
AA	44.07	42.09 ± 00.21	40.66	44.45 ± 01.41	48.43 ± 01.99	48.46 ± 04.17	14.94 ± 00.86	62.01 ± 01.18	44.69 ± 02.31	**62.78 ± 02.31**
$\kappa(\times 100)$	53.86	70.04 ± 00.33	58.57	59.96 ± 02.40	62.16 ± 01.10	75.08 ± 01.35	02.65 ± 03.60	79.88 ± 00.27	58.05 ± 02.01	**85.98 ± 01.27**

TABLE IX

OA, AA, AND KAPPA VALUES ON THE AUGSBURG DATASET (IN %) BY CONSIDERING HS IMAGE AND SAR DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	81.98	87.15 ± 00.77	91.51	81.26 ± 02.95	90.96 ± 01.32	78.21 ± 00.56	87.60 ± 06.19	90.01 ± 00.48	92.04 ± 01.87	**94.65 ± 00.62**
2	86.17	95.82 ± 00.10	86.32	88.32 ± 01.78	82.07 ± 00.74	93.82 ± 00.67	95.26 ± 02.62	94.27 ± 00.78	93.60 ± 02.29	**96.90 ± 00.62**
3	29.66	64.62 ± 02.34	10.03	42.43 ± 14.40	48.31 ± 10.85	63.11 ± 11.45	51.35 ± 72.62	64.62 ± 05.57	43.28 ± 07.08	**69.80 ± 06.43**
4	51.28	84.33 ± 00.80	62.78	67.42 ± 02.97	61.05 ± 04.86	86.24 ± 01.07	91.14 ± 02.98	85.53 ± 00.91	77.08 ± 04.11	**93.98 ± 01.65**
5	36.90	27.72 ± 01.02	22.75	09.88 ± 06.07	18.48 ± 07.96	11.15 ± 08.03	10.52 ± 11.42	29.00 ± 00.63	**45.32 ± 05.54**	32.70 ± 11.81
6	13.80	13.06 ± 02.27	00.00	10.26 ± 05.48	12.11 ± 02.80	09.67 ± 04.92	00.00 ± 00.00	**20.41 ± 01.26**	06.96 ± 01.01	10.52 ± 04.31
7	07.37	11.44 ± 00.39	11.28	09.93 ± 03.22	29.13 ± 01.00	09.42 ± 05.13	24.33 ± 34.41	**44.17 ± 02.06**	29.44 ± 04.77	23.98 ± 07.85
OA	67.29	85.00 ± 00.28	71.65	73.96 ± 01.17	71.79 ± 00.65	83.04 ± 00.47	83.42 ± 01.14	86.10 ± 00.26	81.77 ± 02.30	**90.49 ± 00.20**
AA	43.88	54.88 ± 00.28	40.67	44.21 ± 02.81	48.88 ± 02.61	50.23 ± 01.09	40.75 ± 02.04	**61.14 ± 00.85**	55.39 ± 01.98	60.36 ± 02.36
$\kappa(\times 100)$	54.13	78.62 ± 00.38	58.62	62.20 ± 01.14	59.84 ± 00.63	75.49 ± 00.62	75.26 ± 01.90	80.06 ± 00.33	74.06 ± 03.04	**86.26 ± 00.23**

TABLE X

OA, AA, AND KAPPA VALUES ON THE AUGSBURG DATASET (IN %) BY CONSIDERING HS IMAGE AND DSM DATA

Class No.	Conventional Classifiers			Classical Convolutional Networks				Transformer Networks		
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	RNN	ViT	SpectralFormer	MFT
1	81.99	84.94 ± 00.97	91.51	87.75 ± 00.49	78.92 ± 01.14	77.41 ± 04.03	00.00 ± 00.00	90.39 ± 00.85	90.12 ± 02.42	**94.77 ± 02.29**
2	86.16	88.48 ± 00.06	86.21	93.69 ± 00.22	85.50 ± 01.49	95.47 ± 00.93	**99.92 ± 00.12**	95.04 ± 01.07	86.98 ± 01.18	96.03 ± 01.84
3	29.66	64.50 ± 02.06	10.23	28.96 ± 04.51	56.41 ± 19.63	61.22 ± 06.55	00.00 ± 00.00	59.49 ± 03.02	37.31 ± 03.93	**72.52 ± 10.54**
4	51.29	73.27 ± 00.55	62.80	89.25 ± 00.62	57.72 ± 02.86	86.15 ± 00.07	16.91 ± 23.92	86.34 ± 01.78	58.17 ± 11.82	**93.52 ± 01.96**
5	36.90	26.83 ± 00.59	22.94	18.61 ± 01.86	10.45 ± 04.10	09.31 ± 02.24	00.00 ± 00.00	21.16 ± 14.48	47.74 ± 08.32	**51.31 ± 05.13**
6	13.80	**17.97 ± 01.99**	00.00	15.85 ± 05.68	08.85 ± 08.48	09.22 ± 03.42	00.00 ± 00.00	15.53 ± 11.50	02.32 ± 00.18	13.49 ± 09.44
7	07.37	10.20 ± 00.17	10.75	23.60 ± 01.14	23.65 ± 04.78	09.67 ± 01.79	00.00 ± 00.00	**44.72 ± 01.77**	20.86 ± 03.16	31.28 ± 10.80
OA	67.29	78.04 ± 00.07	71.62	84.43 ± 00.18	70.07 ± 01.11	83.40 ± 00.76	38.80 ± 00.04	86.34 ± 00.40	71.84 ± 03.87	**90.49 ± 00.54**
AA	43.88	52.31 ± 00.17	40.64	51.10 ± 00.61	45.93 ± 02.28	49.78 ± 00.59	14.30 ± 00.02	58.95 ± 03.61	49.07 ± 01.38	**64.70 ± 00.44**
$\kappa(\times 100)$	54.14	68.91 ± 00.09	58.60	77.21 ± 00.26	57.06 ± 01.91	75.90 ± 01.14	06.22 ± 08.80	80.31 ± 00.68	60.15 ± 04.79	**86.35 ± 00.77**

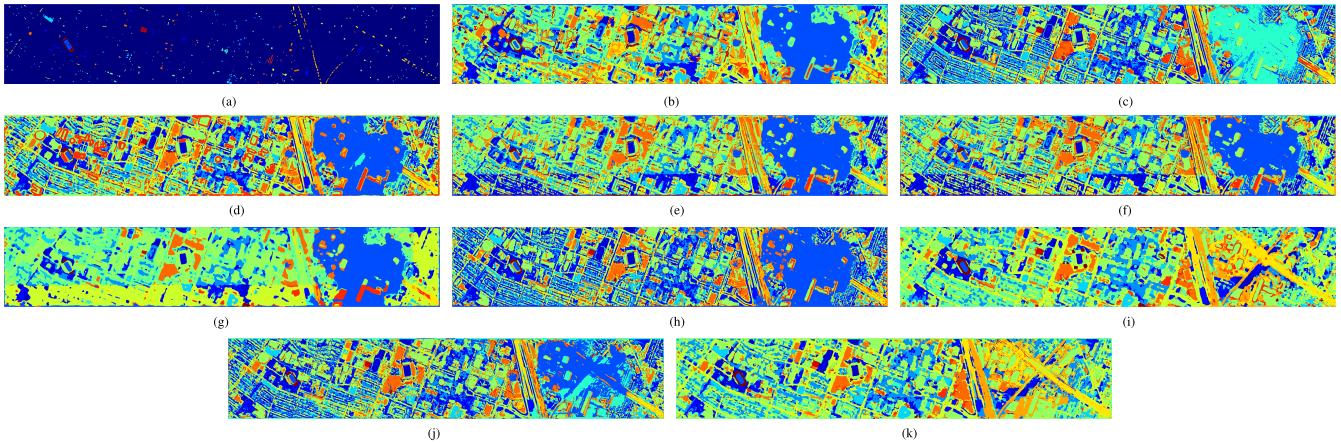


Fig. 8. (a) Ground truth and classification maps obtained for the UH dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) RNN, (i) ViT, (j) SpectralFormer, and (k) MFT using disjoint training samples.

other hand, RF gives higher accuracy using SAR data than DSM data. In the case of classical networks, CNN1D performs better-using DSM, CNN2D and RNN perform better using SAR, and CNN3D performs similarly in both cases.

D. Visual Comparison

We make a qualitative evaluation by visualizing the classification maps obtained by different methods. Figs. 8–13 depict the obtained classification maps for the hyperspectral datasets, i.e., Houston, Trento, MUUFL, and Augsburg, along with their corresponding other modalities, such as LiDAR, MS image, SAR, and DSM data, respectively. Although the conventional classifiers, i.e., KNN, RF, and SVM, produce rich classification maps, the maps still have salt and pepper noise in the boundary regions of the land-cover classes, since they only rely on the spectral information from HS image

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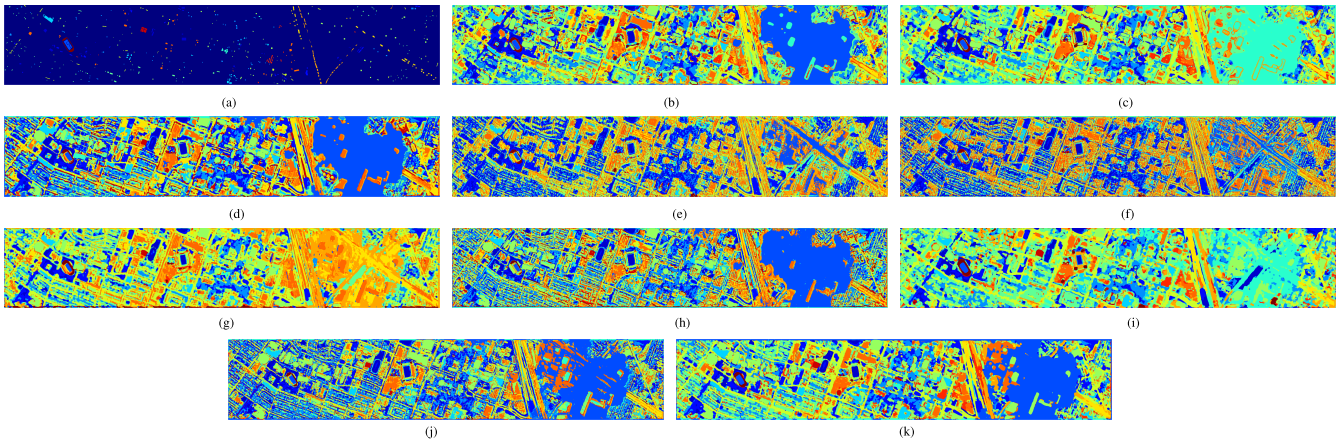


Fig. 9. (a) Ground truth and classification maps obtained by HSI and LiDAR data for the UH dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) RNN, (i) ViT, (j) SpectralFormer, and (k) MFT using disjoint training samples.

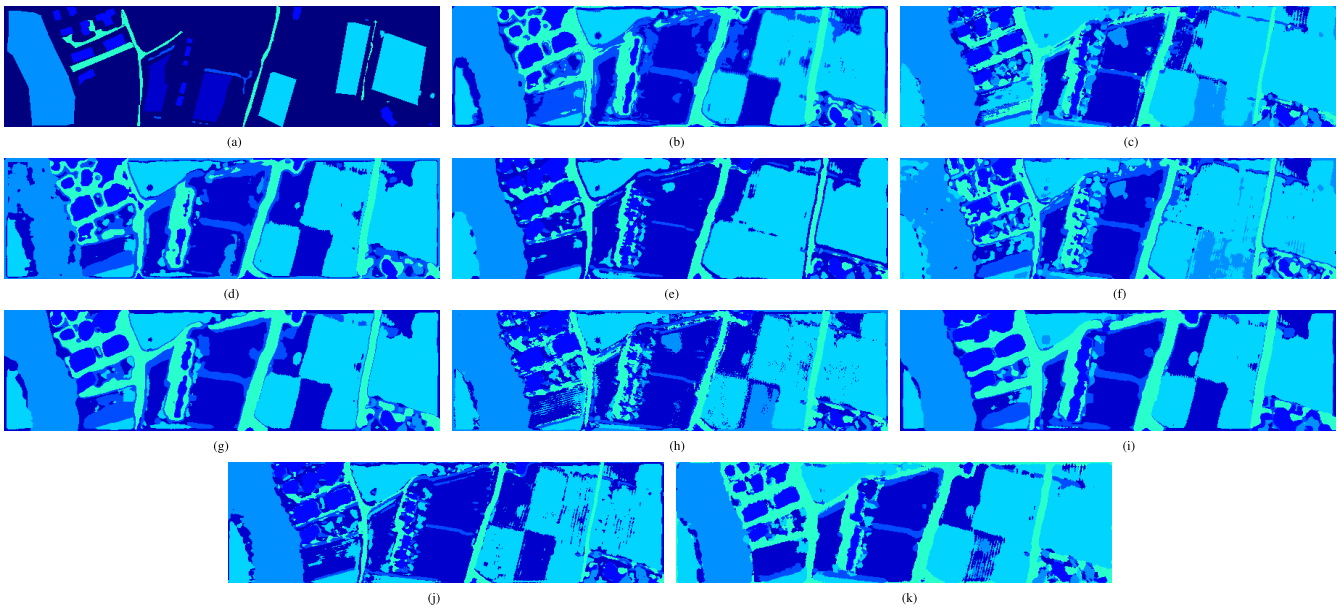


Fig. 10. (a) Ground truth and classification maps obtained by HSI and LiDAR data for the Trento dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) RNN, (i) ViT, (j) SpectralFormer, and (k) MFT using disjoint training samples.

TABLE XI

OA, AA, AND KAPPA VALUES ACHIEVED BY MFT WITH DIFFERENT TOKENIZATION, I.E., PT AND CT, FOR UH (HSI + LIDAR), UT (HSU + LIDAR), MUUFL (HSI + LIDAR), UH (HSI + MS), AUGSBURG (HSI + SAR), AND AUGSBURG (HSI + DSM) DATASETS

Class No.	Houston (HSI+LiDAR)		Trento (HSI+LiDAR)		MUUFL (HSI+LiDAR)		Houston (HSI+MS)		Augsburg (HSI+SAR)		Augsburg (HSI+DSM)	
	PT	CT	PT	CT	PT	CT	PT	CT	PT	CT	PT	CT
1	82.34 ± 00.47	**84.90 ± 03.02**	**98.71 ± 00.80**	98.23 ± 00.38	97.52 ± 00.38	**97.90 ± 00.39**	82.72 ± 00.41	**87.75 ± 07.12**	94.65 ± 00.62	**95.53 ± 00.69**	**95.38 ± 00.38**	94.77 ± 02.29
2	**88.78 ± 05.27**	84.71 ± 00.56	**99.50 ± 00.12**	99.34 ± 00.02	**92.35 ± 02.13**	92.11 ± 01.58	85.09 ± 00.09	**85.15 ± 00.00**	**96.90 ± 00.62**	96.81 ± 00.91	**97.21 ± 00.09**	96.03 ± 01.84
3	98.15 ± 01.08	**98.61 ± 01.01**	80.12 ± 07.78	**89.84 ± 09.00**	**93.21 ± 01.30**	91.80 ± 00.82	**98.55 ± 00.76**	98.42 ± 00.32	**69.80 ± 06.43**	66.98 ± 02.92	55.10 ± 07.22	**72.52 ± 10.54**
4	94.35 ± 02.40	**96.88 ± 01.45**	**99.96 ± 00.03**	99.82 ± 00.26	**93.37 ± 01.85**	91.59 ± 02.25	**95.99 ± 02.39**	92.90 ± 02.89	93.98 ± 01.65	**94.31 ± 01.81**	92.42 ± 00.34	**93.52 ± 01.96**
5	99.12 ± 01.25	**99.97 ± 00.04**	99.82 ± 00.23	**99.93 ± 00.05**	95.18 ± 00.13	**95.60 ± 01.21**	**99.78 ± 00.09**	99.21 ± 00.39	**32.70 ± 11.81**	24.54 ± 06.34	43.28 ± 11.66	**51.31 ± 05.13**
6	**99.30 ± 00.00**	90.44 ± 07.58	84.29 ± 01.40	**88.72 ± 00.94**	84.65 ± 02.48	**88.19 ± 03.49**	97.20 ± 01.98	**97.67 ± 01.74**	10.52 ± 04.31	**12.31 ± 02.19**	**14.06 ± 04.71**	13.49 ± 09.44
7	**88.56 ± 01.16**	81.84 ± 05.59			89.93 ± 02.23	**90.27 ± 02.13**	86.32 ± 00.69	**87.28 ± 01.38**	**23.98 ± 07.85**	15.28 ± 07.14	16.46 ± 04.41	**31.28 ± 10.80**
8	86.89 ± 05.33	**89.78 ± 01.83**			97.24 ± 00.48	**97.26 ± 00.53**	**81.16 ± 08.04**	77.71 ± 00.98				
9	87.91 ± 03.90	**91.94 ± 01.20**			58.89 ± 05.71	**61.35 ± 03.80**	87.76 ± 00.74	**88.29 ± 01.72**				
10	64.70 ± 00.92	**65.03 ± 12.68**			11.69 ± 04.51	**17.43 ± 04.63**	**74.71 ± 15.98**	65.67 ± 02.15				
11	98.64 ± 00.70	**99.81 ± 00.20**			**74.87 ± 04.06**	72.79 ± 09.25	93.71 ± 06.01	**96.90 ± 00.47**				
12	**94.24 ± 02.19**	89.95 ± 01.93					96.16 ± 00.87	**97.05 ± 00.99**				
13	**90.29 ± 00.66**	88.42 ± 03.49					**92.51 ± 00.17**	91.23 ± 00.76				
14	**99.73 ± 00.38**	99.73 ± 00.38					**100.0 ± 00.00**	99.46 ± 00.76				
15	**99.58 ± 00.17**	94.08 ± 07.07					86.82 ± 08.99	**95.07 ± 02.46**				
OA	**89.80 ± 00.53**	89.39 ± 00.78	97.82 ± 00.24	**98.32 ± 00.25**	94.26 ± 00.20	**94.34 ± 00.07**	**89.15 ± 00.96**	88.97 ± 00.36	**90.49 ± 00.20**	90.40 ± 00.17	89.48 ± 00.27	**90.49 ± 00.54**
AA	**91.51 ± 00.40**	90.41 ± 01.10	93.73 ± 01.52	**95.98 ± 01.64**	80.81 ± 01.02	**81.48 ± 00.70**	90.56 ± 00.93	**90.65 ± 00.29**	**60.36 ± 02.36**	57.97 ± 02.33	59.13 ± 02.80	**64.70 ± 00.44**
$\kappa(\times 100)$	**88.93 ± 00.59**	88.48 ± 00.85	97.08 ± 00.33	**97.75 ± 00.33**	92.41 ± 00.27	**92.51 ± 00.10**	**88.22 ± 01.04**	88.02 ± 00.39	**86.26 ± 00.23**	86.07 ± 00.28	84.80 ± 00.43	**86.35 ± 00.77**

data. The imperfections in the generated classification maps can be slightly reduced by considering the complementary information from other sources of modalities. However, the DL models successfully capture the nonlinear relationship between the input and output feature maps due to their learnable feature extraction capabilities. Hence, CNN1D, CNN2D, and

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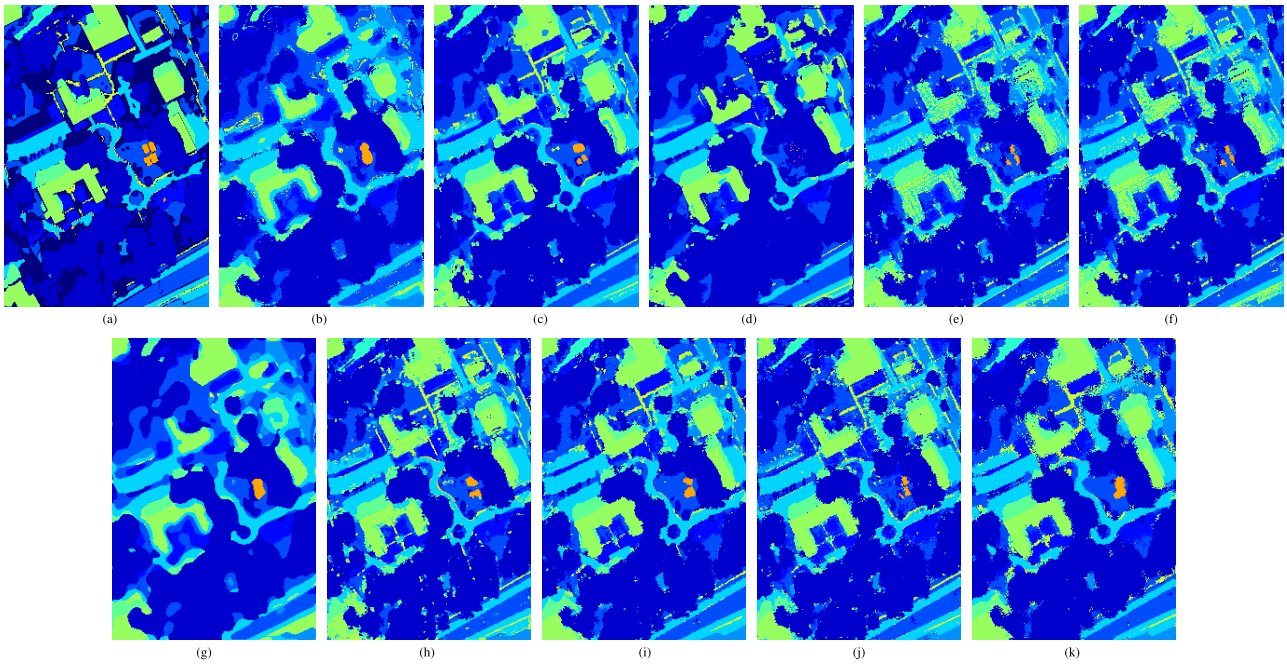


Fig. 11. (a) Ground truth and classification maps obtained by HS image and LiDAR data for the MUUFL dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) RNN, (i) ViT, (j) SpectralFormer, and (k) MFT using disjoint training samples.

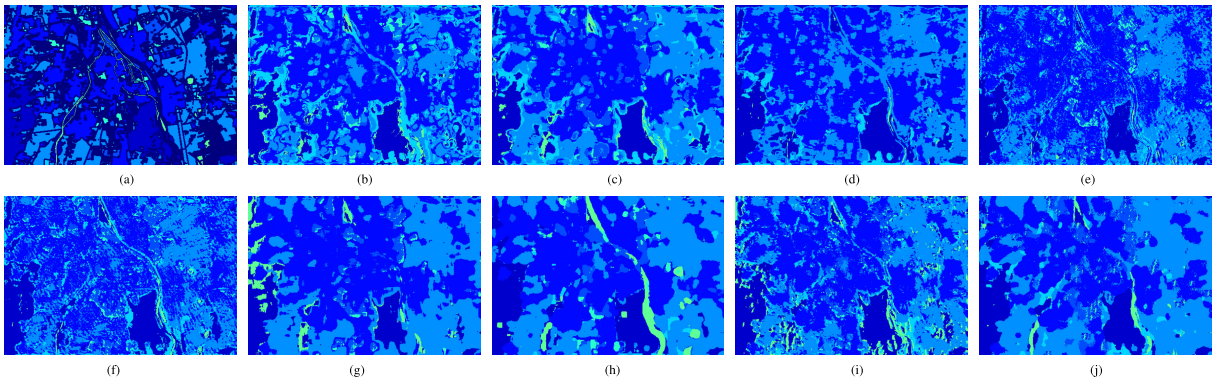


Fig. 12. (a) Ground truth and classification maps obtained by HSI and SAR data for the Augsburg dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) ViT, (i) SpectralFormer, and (j) MFT using disjoint training samples.

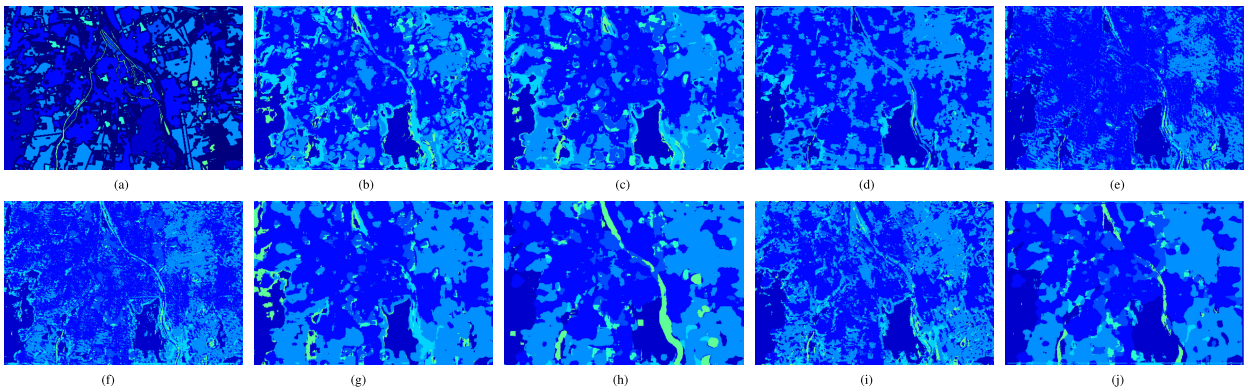


Fig. 13. (a) Ground truth and classification maps obtained by HSI and DSM data for the Augsburg dataset by (b) KNN, (c) RF, (d) SVM, (e) CNN1D, (f) CNN2D, (g) CNN3D, (h) ViT, (i) SpectralFormer, and (j) MFT using disjoint training samples.

CNN3D produce relatively smooth classification maps where land-use and land-cover class boundaries are well separable. ViT was found effective for HS classification due to its ability to extract high-level sequential abstract representations from HS data resulting. Therefore, classification maps are of better visual quality compared with classical networks. By enhancing

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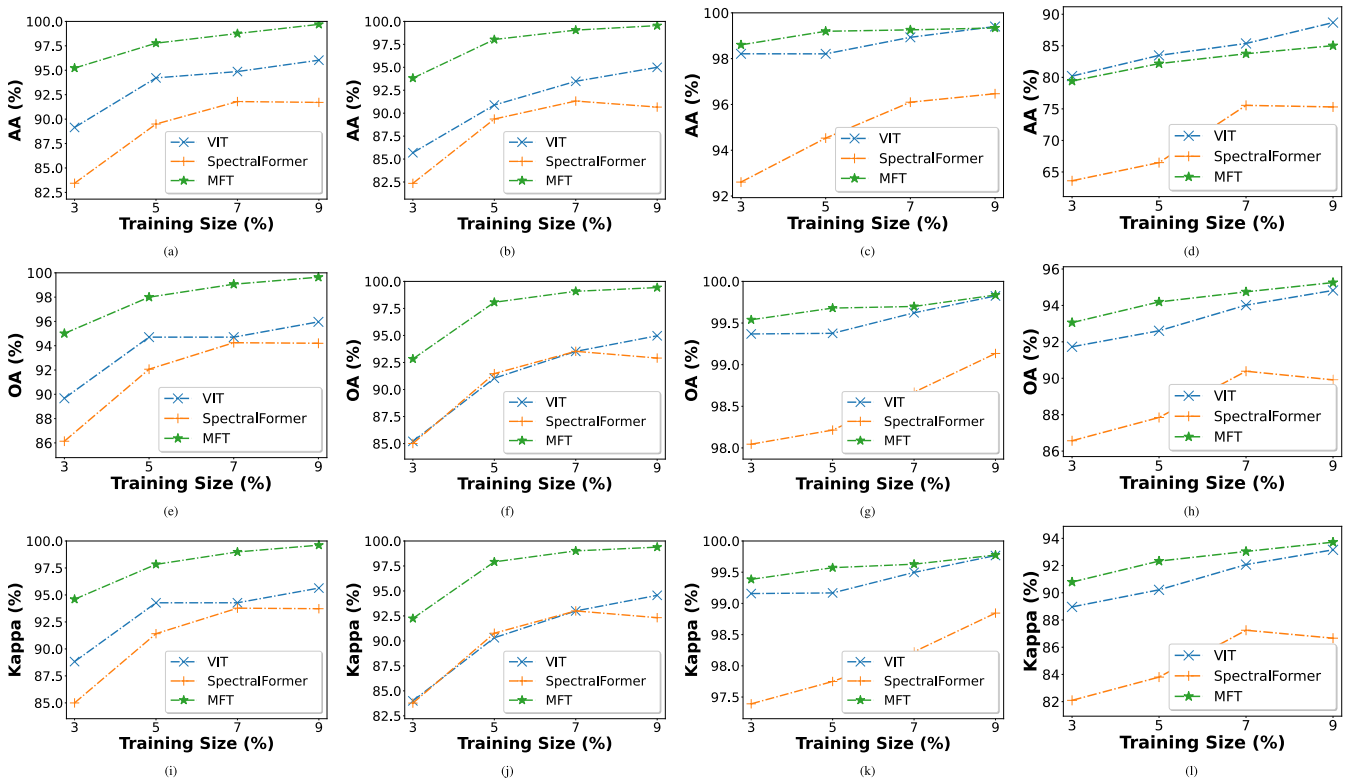


Fig. 14. AA, OA, and kappa coefficient (κ) achieved by different methods with varying training sample sizes, which are randomly taken from (a), (e), and (i) UH (HSI + LiDAR), (b), (f), and (j) UH (HSI + MS), (c), (g), and (k) Trento (HSI + LiDAR), and (d), (h), and (l) MUUFL (HSI + LiDAR) datasets.

the neighboring spectral information and conveying positional information across layers more effectively, the proposed MFT obtains highly desirable classification maps, especially in terms of texture and edge details compared with ViT and SpectralFormer. As observed from the generated classification maps, distinct class boundaries and remarkable improvements in the visual quality are obtained by considering the other sources of modalities during feature extraction and learning of all the networks. More noticeable improvements can be seen for the proposed MFT network due to the better exchange of information between HS patch and other modalities. The learning of complementary information helps find strong nonlinear sequential relations from the HS image data. Hence, the generated classification maps contain more realistic and finer details inside the land-use and land-cover region.

E. Performance Over Varying Training Ratio

Here, we evaluate the performance of the proposed model with respect to the number of labeled training samples. In the case of inadequate or unreliable training samples, underfitting (also called the Hughes phenomena) or overfitting (nonreliable) issues can occur. Therefore, selecting the appropriate number of training samples is another crucial factor for classification performance. The purpose of this section is to analyze the experimental results on different percentages of training samples where 3%, 5%, 7%, and 9% training samples are randomly selected to train the models (both the one presented in this article and the comparative methods), and the remaining samples are used for testing. Other parameters remain the same, as discussed in Section IV-B.

Fig. 14 compares the detailed classification performances of the transformer models using AA [Fig. 14(a)–(d)], OA [Fig. 14(e)–(h)], and κ [Fig. 14(i)–(l)] metrics, respectively, on four datasets, i.e., Houston (HSI + LiDAR), Trento (HSI + LiDAR), MUUFL (HSI + LiDAR), and Houston (HSI + MS), with different percentages (3%, 5%, 7%, and 9%) of randomly selected training samples. The y-axis of the graphs shows the metrics (OA, AA, or κ), while the x-axis shows the training sample size in percent. Each model is denoted with a different marker and color, which are consistent throughout Fig. 14. The proposed MFT is shown in green.

The graphs illustrate that the experimental results obtained with 3% training samples are inferior to those obtained with 9% training samples for all four datasets. In Houston (HSI + LiDAR) and Houston (HSI + MS), the proposed MFT performs approximately 4% and 6% better than the second-best model, i.e., ViT, respectively, in all three accuracies for all percentages of training samples. In other datasets, though the proposed MFT performs better than ViT, the improvement is not much for larger training sample sizes. It is quite evident that even with the decrease in training samples, the performance of the proposed MFT is significantly better than the other transformer networks.

F. Model Stability Analysis

1) Ablation Study: Regardless of the source of multimodal data, the dimension to tokenize plays a crucial role in the final classification performance and should be fine-tuned in addition to learning parameters and hyperparameters. Therefore, it is

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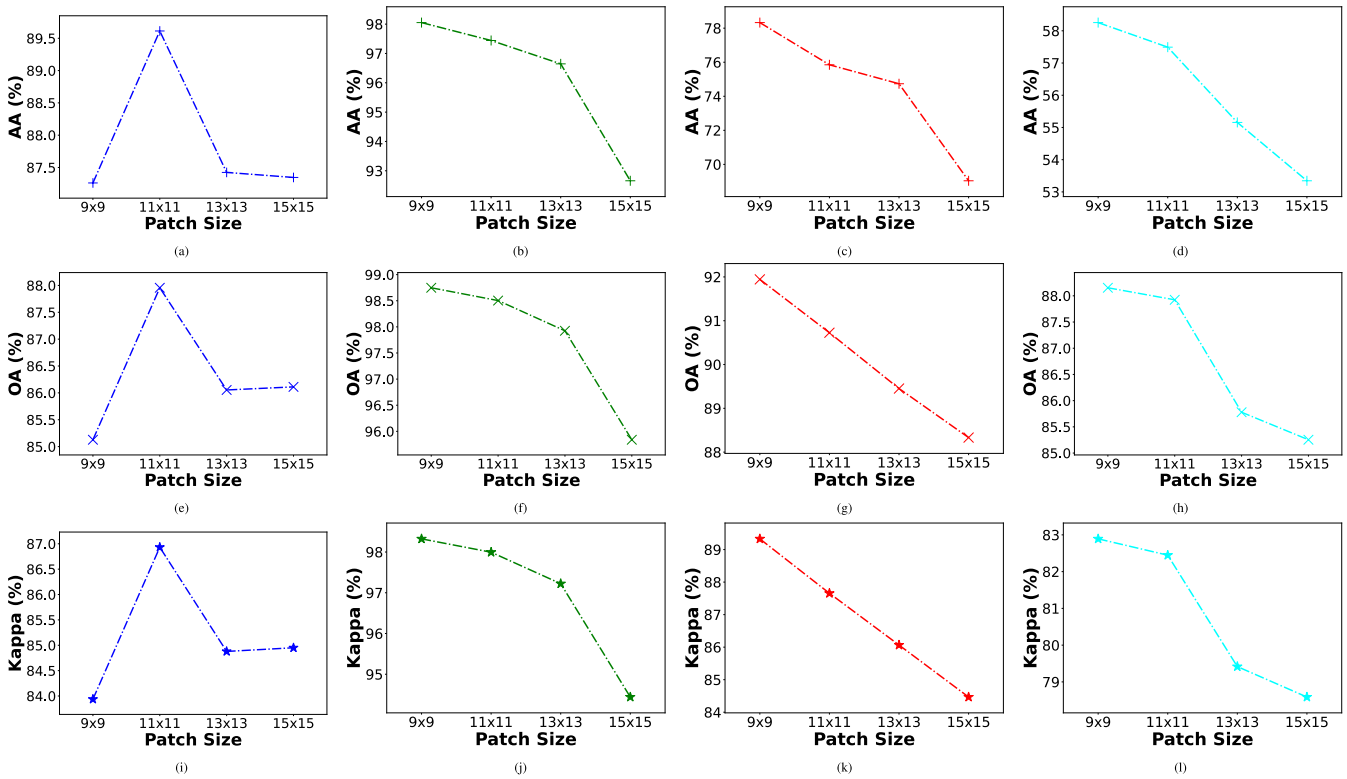


Fig. 15. AA, OA, and kappa coefficient (κ) achieved by the MFT network with varying patch sizes, which are taken from (a), (e), and (i) UH (HSI + LiDAR), (b), (f), and (j) Trento (HSI + LiDAR), (c), (g), and (k) MUUFL (HSI + LiDAR), and (d), (h), and (l) Augsburg (HSI + SAR) datasets.

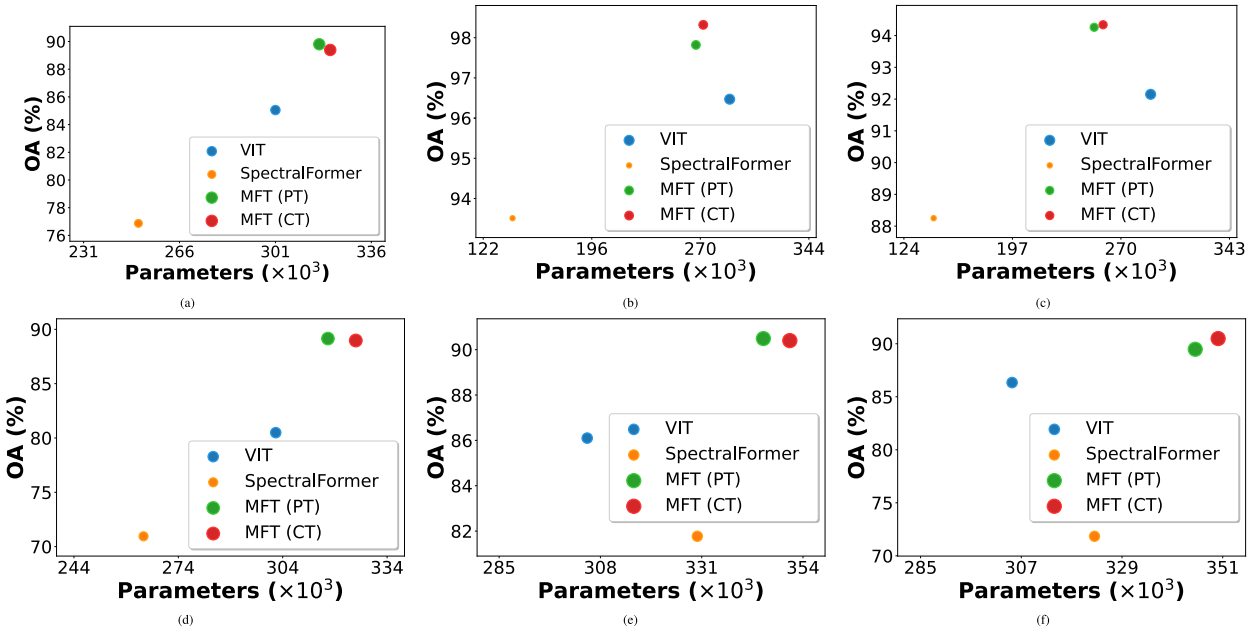


Fig. 16. Comparison of the transformer networks in terms of overall classification accuracy, network parameters, and FLOPs, indicated by radii of circles taken from (a) UH (HSI + LiDAR), (b) UT (HSI + LiDAR), (c) MUUFL (HSI + LiDAR), (d) UH (HSI + MS), (e) Augsburg (HSI + SAR), and (f) Augsburg (HSI + DSM) datasets.

essential to compare the model's performance with respect to pixel tokenization and channel tokenization. Hence, we investigate the classification performance of both variations on all four datasets, along with their respective combined multimodal data. Table XI shows the comparison of the classification accuracies of both methods in terms of OA, AA, and κ for all the datasets. The optimal values are marked in bold. Overall, the performances of both methods are quite similar, but channel tokenization narrowly exceeds pixel tokenization. The reason may be that the channels represent the features better than pixels after convolution operations.

2) Impact of Patch Size: The performance of the proposed MFT network is also influenced by the spatial size of the input HSI patches. As shown in Fig. 15, the best and more stable AAs, OAs, and Kappa values are achieved by the proposed model using a window of size (11 × 11) for all the experimentally evaluated datasets. In all these experiments, disjointed train/test samples are utilized, as explained in Section IV-A.

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It can also be observed from the figures that, even the higher values in terms of AAs, OAs, and Kappa values can be achieved over the University of Trento (UT) (H + L), MUUFL (H + L), and Augsburg (H + S) datasets using a patch size of (9 × 9), but the performance is not satisfactory for the UH (H + L) multimodalities. The results for patches of size 13 × 13 and 15 × 15 show that performance can significantly be reduced when increasing the patch size. It is worth mentioning that increasing the spatial window size will lead to an increase in redundancy. Hence, we have considered 11 × 11 HSI patches for the entire experimental evaluation.

3) Hyperparameter Sensitivity Analysis: The proposed MFT is not only effective but also fairly efficient in terms of computational complexity. The parameters and computations of the proposed MFT against other transformer networks are compared in Fig. 16(a)–(f). They illustrate the overall accuracies, parameters, and computations (FLOPs) for UH (HSI + LiDAR), UT (HSI + LiDAR), MUUFL (HSI + LiDAR), UH (HSI + MS), Augsburg (HSI + SAR), and Augsburg (HSI + DSM) datasets, respectively. The radii of circles indicate the computations (FLOPs). The efficiency of the proposed MFT is quite evident in Trento (HSI + LiDAR) and MUUFL (HSI + LiDAR), where it uses fewer parameters and FLOPS than ViT (the model with second-best accuracies). In other cases, though, the parameters and FLOPS of the proposed MFT are higher than others, the increase in performance makes up for it. As we can see with Houston (H + MS), the proposed MFT exhibits an impressive 8.66% increase in OA than the next best model, i.e., ViT. Such a massive gain in classification accuracy makes the parameter trade-off justified.

V. Conclusion

In this article, we propose a new MFT network based on an `mCrossPA` module to fuse HSI and other sources of multimodal data for land-cover classification. Instead of using conventional feature fusion techniques, other sources of multimodal data are used as an external classification (`CLS`) token that is incorporated into the `mCrossPA` block in the transformer encoder, which helps learning long-range feature dependencies. As the `CLS` token captures information that is complementary to that of the HSI data, it also helps in achieving a better generalization of the model and improves classification accuracy. The `CLS` token can be generated from multiple sources of easily available data, such as MSIs, SAR, DSM, and LiDAR. As a result, the model's performance is significantly increased with minimal effort.

Fusing multimodal data (such as HSIs, MSIs, SAR, DSMs, and LiDAR) using recently developed transformers are challenging and require significant effort. However, the extensive experimental results presented in this work confirm that the proposed MFT can successfully fuse multimodal data using transformers, which are inherently better at classification tasks than classical convolutional models. The proposed MFT performs better than all the other tested models with all the considered datasets. Therefore, we believe that the proposed MFT exhibits the capacity to perform multimodal fusion tasks easily in RS and EO due to its superior fusion capabilities.

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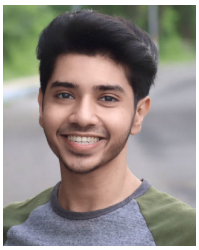
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